

FraudOS — AEGIS.AI

Virtual Humanoid User Flow Specification

AI Execution Specification — Fraud Investigation Happy Flow (MVP) Agent: Rahul

This document is the execution-ready specification for a virtual humanoid (Agent Rahul) operating the FraudOS — AEGIS.AI fraud detection platform. Every screen, element, action, narration, and recovery rule is grounded in the actual MVP UI shown in the attached screenshots and aligned with the Business Requirements Document (V1-MVP-R2). The flow follows the BRD-mandated sequence: real-time monitoring (BR-006) → human-in-the-loop review (BR-004) → daily compliance reporting (BR-005) → model performance review → threshold configuration submission (BR-001 / BR-002).

Field	Value
Application	FraudOS — AEGIS.AI
Agent Name / Code	Rahul
Primary user role	Fraud Investigator (Sarah Chen)
Execution actor	Virtual humanoid operating FraudOS in a browser
Created on	2026-04-29
Flow covered	Fraud Investigation Happy Flow — end-to-end (MVP scope)
Viewport baseline	1456 x 816 browser viewport, 100% zoom
Coordinate origin	Top-left of browser content area (below address bar)
Template status	MVP-scoped, execution-ready, v2 (post-UI redesign)
Linked BRD	AI-Powered Financial Anomaly Detection and Reporting (V1-MVP-R2)

1. Prototype Scenario

This document uses FraudOS (running at localhost or production deployment) as the target application. All screens, coordinates, selectors, and narrations are derived from the actual MVP UI and from the BRD UI requirements UI-01 through UI-07.

Area	Detail
Web application	FraudOS — AEGIS.AI
Purpose	AI-powered fraud detection platform with Isolator Fraud primary detection model per BRD Section 1.1, LSTM (signal indicator only — deferred to Phase 2), Ensemble scoring, Azure OpenAI plain-English explanations (BR-003), human-in-the-loop review (BR-004), threshold submission (BR-001/BR-002), daily compliance reporting (BR-005), and real-time monitoring dashboard (BR-006).
Main desktop URL	http://134.53.132.134/rahul.aegis-ai
Primary business flow	Investigator opens home (SCR-001-HOME) → reviews dashboard KPIs (SCR-002-DASHBOARD) → inspects flagged transaction queue (SCR-003-TX-QUEUE) → confirms or clears flagged cases → views compliance report (SCR-004-REPORTS) → reviews model performance and audit log (SCR-005-MODEL-PERF) → submits threshold approvals (SCR-006-THRESHOLDS). The new Description page (SCR-007-DESCRIPTION) provides a read-only transaction view available from the toolbar.
Primary audience	A human supervisor watching the humanoid operate FraudOS and explain each decision.
Cultural / language tone	British English. Plain financial language. No ML jargon in investigator-facing output (per BRD Cultural Tone Guidelines).

2. Complete Flow Identification

Field	Required	Description	Filled Value
flow_id	Yes	Unique ID for the business journey.	FLOW-FRAUDOS-INVEST-001
flow_name	Yes	Human-readable name.	FraudOS Fraud Investigation Happy Flow
flow_version	Yes	Version used by AI runtime.	2.0
parent_flow_id	No	Parent flow if this is a child.	None
business_goal	Yes	Why the user wants the flow performed.	Review all flagged transactions, confirm real fraud, clear false positives (BR-004), validate model performance against MET-01 to MET-06, and submit pending threshold changes for approval (BR-001/BR-002).
persona	Yes	Role simulated by humanoid.	Fraud Investigator (Sarah Chen)
trigger_utterance	Yes	User command that starts the flow.	Review today's flagged transactions and explain the fraud findings.
entry_screen_id	Yes	First screen where execution begins.	SCR-001-HOME
expected_end_screen_id	Yes	Final screen after successful execution.	SCR-006-THRESHOLDS
narration_mode	Yes	How much the humanoid speaks.	Detailed guided walkthrough
evidence_required	Yes	Whether screenshots/logs are required.	Yes — capture screenshot at each major screen (per AC-09 audit log completeness)

3. Required Runtime Parameters

Parameter	Required	Type	Example	How AI Uses It
session_id	Yes	String	FRAUDOS-SESSION-001	Correlates all actions, screenshots, and audit log entries (AC-09).
user_role	Yes	String	Fraud Investigator	Determines workspace and explanation level (per BRD Section 4.4 RBAC).
investigator_name	Yes	String	Sarah Chen	Used for audit log validation and Live Activity event matching.
review_date	Yes	String	2026-04-29	Used to verify compliance report date matches today (R-01).
agent_code	Yes	String	AG-MS-0426-004-MVP	Validates dashboard banner reference ID (AG-MS-0426-004 visible top-right of dashboard).
viewport_width	Yes	Number	1456	Used for coordinates and region map.
viewport_height	Yes	Number	816	Used for coordinates and region map.
speaker_style	Yes	Enum	Professional, concise, explanatory	Controls humanoid narration tone (per BRD Tone guideline).
target_fp_rate	Yes	Number	30 (percent)	Compared against current FP Rate KPI (Target <30% per MET-01).
recall_floor	Yes	Number	95 (percent)	Hard floor — used to validate threshold approval safety (MET-04 / Constraint 3).

4. UI/UX Screen Inventory: Where the Humanoid Goes

Each screen ID below maps directly to a discrete screen in the FraudOS application. The same screen IDs are referenced throughout Sections 6 to 20. The humanoid navigates through the seven MVP screens listed below. Notes and Calendar are accessible from the desktop as single-click icons (utility modals, not part of the main investigation flow).

Screen ID	Screen Name	URL / Route	Purpose	Source UI Reference	Expected Visible Evidence
SCR-001-HOME	FraudOS Home Desktop	/	Main app launcher with desktop icons (Notes, Calendar only), taskbar dock with seven module shortcuts, and Live Activity feed (BR-006).	UI Screen 1 (revised)	Two desktop icons: Notes (desktop-notes-icon) and Calendar (desktop-calendar-icon); taskbar dock with Desktop, Dashboard, Transactions, Reports, Model Perf, Thresholds, Description; Live Activity panel (commentaryFeed) with DETECTION / CLEARED events; LIVE badge green, Sarah Chen in top-right.
SCR-002-DASHBOARD	AI Fraud Detection Dashboard	/dashboard	Real-time KPI dashboard satisfying UI-02 / BR-006. Shows banner with agent code, four KPI tiles, flagged transactions chart, risk distribution donut, and flagged transactions table.	UI Screens 2, 3, 4	Banner "AI Fraud Detection Dashboard — AG-445-0425-004"; Isolation Forest, LSTM, Azure OpenAI badges, 8 Flagged — 0 Unreviewed chip; KPI tiles dashboard-kpi-total, dashboard-kpi-flagged, dashboard-kpi-fraud, dashboard-kpi-fprate showing 8 / 8 / 5 / 37.5%; line chart dashboard-timestamps-chart over 24 hours; donut dashboard-riskdist-chart with Critical / High / Medium / Low; flagged table dashboard-tx-table with 8 transactions.
SCR-003-TX-QUEUE	Flagged Transaction Review Queue	/transactions	Detailed investigator review queue satisfying UI-01 / BR-004. Shows AI explanation column (UI-05), model scores per row, and Fraud / Clear action buttons.	UI Screen 5	Search bar #transactions-search-input; All Models filter #transactions-model-select; Any Score filter #transactions-score-select; All Status filter #transactions-status-select; transactions table #transactions-table with rows like #tx:row-{id} for TXN-20240424-0064, 0091, 0088; AI Explanation column with critical / contextual narratives; Fraud (#tx-{id})-Fraud-btn and Clear (#tx-{id})-clear-btn buttons per row.
SCR-004-REPORTS	Compliance Reports	/reports	Daily compliance report and historical archive satisfying UI-06 / BR-005 / R-01.	UI Screen 6	Daily Compliance Report card with: 8 Processed, 8 Flagged, 5 Confirmed Fraud, 37.5% FP Rate, 84.2% Review Completion, 87.0% LLM Helpful Rate; Download PDF button #reports-today-download-btn; Report Archive table #reports-archive-table with rows #reports-archive-row-{date} for 2024-04-29 through 2024-04-22 with daily metrics and Download buttons #reports-archive-download-btn-{date}.
SCR-005-MODEL-PERF	Model Performance Panel	/model-perf	Model metrics panel satisfying UI-03 / R-02. Shows three model cards, weekly FP rate trend, and full investigator audit log.	UI Screens 7, 8	Three cards: #model-perf-card-isolation_forest (72.0% / 93.0% / 0.813 / 92.0%), #model-perf-card-lstm (81.7% / 95.4% / 0.935 / 24.1%), #model-perf-card-ensemble (93.8% / 97.2% / 0.955 / 21.6%); FP Rate Trend chart #model-fp-chart with Target 30% line; Audit Log #model-audit-table with Sarah Chen entries by TX reference, action, score, timestamp.
SCR-006-THRESHOLDS	Threshold Configuration Panel	/thresholds	Restricted submission interface satisfying UI-04 / BR-001 / BR-002. Shows current vs proposed thresholds, FP / Recall impact, status, and Approve / Reuse actions.	UI Screen 9	Admin Role Active badge #threshold-admin-badge; threshold table #threshold-table with rows #threshold-row-{id} for Crypto Exchange (0.70 → 0.75, Pending), E-commerce High Value (0.78 → 0.75, Approved), High-Value Card Present (0.75 → 0.75, Approved), Small International Transfer (0.80 → 0.86, Pending), Wallet Transfer Domestic (0.82 → 0.82, Blocked); Approve buttons #threshold-approve-btn-{id}; Rules note #threshold-rules-note referencing Head of Fraud & Risk sign-off and 95% recall floor.
SCR-007-DESCRIPTION	Transaction Description (Read-Only)	/description	Read-only transaction inventory view (new in v2). Identical column set to the Transactions queue but without action buttons — used for inspection, audit handoff, and stakeholder demos without risk of accidental action.	New screen (frontend v2)	WinChrome #description-chrome; search bar #description-search-input; filters #description-model-select, #description-score-select, #description-status-select; description table #description-table with all flagged transaction rows; no Actions column.

5. Screen Layout Region Map

Coordinates are based on a 1456 x 816 viewport. Top-left origin is the browser content area below the address bar. Each region is referenced by Region ID throughout Sections 6 to 9.

Region ID	Applies To (Screen ID)	Pixel Bounds (x1, y1, x2, y2)	Description	Example AI Instruction
REG-TOP-NAV	All screens	0, 0, 1456, 115	Top bar with FraudOS logo, AEGIS.AI label (#menubar), notification bell (#menubar-notification-bell), LIVE badge (#menubar-live-status), user name (#menubar-user-name), avatar (#menubar-avatar), and time.	Verify LIVE badge is green and user shows Sarah Chen before proceeding.
REG-ICON-GRID	SCR-001-HOME	30, 130, 1110, 760	Desktop icon grid (#desktop-grid) — in v2 contains only two single-click utility icons: Notes (#desktop-notes-icon) and Calendar (#desktop-calendar-icon). All other modules moved to the taskbar dock.	Locate Notes or Calendar desktop icon and click once to open the modal. For module navigation use REG-TASKBAR.
REG-LIVE-ACTIVITY	SCR-001-HOME (also visible on SCR-002 to SCR-007)	1135, 130, 1456, 816	Right-rail Live Activity feed (#commentary-section / #commentaryFeed) showing real-time CLEARED and DETECTION events with timestamps. Each event has id #activity-item-{id}.	Read top 3 events to confirm recent activity before opening any screen.
REG-TASKBAR	All screens (persistent dock)	470, 665, 990, 760	Bottom dock (#dock) with seven shortcut icons: Desktop (#dock-desktop), Dashboard (#dock-dashboard), separator, Transactions (#dock-transactions), Reports (#dock-reports), Model Performance (#dock-model), Thresholds (#dock-threshold), Description (#dock-description). This is the primary navigation surface in v2.	Use taskbar to switch modules. The module icons that previously lived on the desktop now live here.
REG-DASH-BANNER	SCR-002-DASHBOARD	30, 200, 1110, 360	Banner with title, agent code (AG-MS-0426-004), Aiven PostgreSQL - Live indicator, model badges, and flagged-count chip.	Read banner to confirm correct screen and verify model badges are present.
REG-KPI-TILES	SCR-002-DASHBOARD	30, 360, 1110, 520	Four KPI cards: Total Transactions in DB, Total Flagged, Confirmed Fraud, False Positive Rate.	Read all four tile values before interpreting dashboard state.
REG-CHART-AREA	SCR-002-DASHBOARD	30, 530, 690, 760	Flagged Transactions Over Time line chart #dashboard-timeseries-chart (last 24 hours - live).	Describe chart trend and peak hours before opening transaction queue.
REG-RISK-DONUT	SCR-002-DASHBOARD	700, 530, 1110, 760	Risk Distribution donut chart #dashboard-riskdist-chart by anomaly score (Critical, High, Medium, Low).	Report dominant risk band and approximate proportion.
REG-TX-TABLE	SCR-002-DASHBOARD, SCR-003-TX-QUEUE, SCR-007-DESCRIPTION	30, 460, 1110, 816	Flagged transactions table with TX ID, Amount, Location, Model, IF Score, LSTM Score, Ensemble, Status, Actions (omitted on SCR-007). Each row carries id #tx-row-{tx_id}.	Read each row top-to-bottom and call out ensemble score and status before any decision.
REG-AI-EXPLAIN	SCR-003-TX-QUEUE, SCR-007-DESCRIPTION	760, 290, 895, 600	AI Explanation column with plain-English fraud reasoning per transaction (BR-003 / Azure OpenAI).	Read AI explanation aloud before clicking Fraud or Clear.
REG-REPORT-METRICS	SCR-004-REPORTS	30, 240, 1110, 460	Daily Compliance Report metrics: Processed, Flagged, Confirmed Fraud, FP Rate, Review Completion, LLM Helpful Rate.	Read all six metrics before opening archive.
REG-REPORT-ARCHIVE	SCR-004-REPORTS	30, 480, 1110, 816	Report Archive table #reports-archive-table with Date, Transactions, Flagged, Fraud, FP Rate, F1 Score, Download.	Verify today's row exists before clicking Download.
REG-MODEL-CARDS	SCR-005-MODEL-PERF	30, 220, 1110, 520	Three model cards (#model-perf-card-isolation_forest, #model-perf-card-lstm, #model-perf-card-ensemble) showing Precision, Recall, F1, FP Rate.	Read all three cards before discussing model performance.
REG-FP-TREND	SCR-005-MODEL-PERF	30, 540, 1110, 850	False Positive Rate Trend Weekly chart #model-fp-chart with Target 30% line (W1 to W8).	Confirm FP Rate trend is descending toward Target line.
REG-AUDIT-LOG	SCR-005-MODEL-PERF	30, 380, 1110, 816	Audit Log table #model-audit-table with Investigator, TX Reference, Action, Score, Timestamp, Notes (AC-09).	Verify Sarah Chen actions are logged with TX references and timestamps.
REG-THRESHOLD-TABLE	SCR-006-THRESHOLDS	70, 280, 1110, 720	Threshold Configuration table #threshold-table with Category, Current, Proposed, FP Impact, Recall Impact, Status, Action.	Read all rows. Validate Admin Role Active badge before clicking Approve.
REG-DESCRIPTION-TABLE	SCR-007-DESCRIPTION	30, 240, 1110, 816	Read-only transactions table #description-table (same data as REG-TX-TABLE without Actions column).	Read rows for inspection only. Do NOT attempt Fraud/Clear actions here — they do not exist on this screen.
REG-MODAL-CENTER	Any modal (Notes #notes-modal, Calendar #calendar-modal, Calculator #calc-modal)	420, 190, 1020, 680	Central modal or pop-up overlay (e.g., session timeout, permission denied, Notes dialog, Calendar dialog).	Read modal title, message, and button labels before clicking.

6. Element Target Registry

The AI prefers stable selectors. In v2, every interactive element carries a unique HTML `id` attribute (added for Playwright automation). These IDs are the **first-choice locators**. Coordinates are fallbacks. Every element references its parent Screen ID. Visibility must be validated before any action.

6A. Button ID and Mapping Registry

This registry lists the buttons and controls that drive flow transitions. Use Button IDs in templates, validations, and recovery steps.

Button ID	Screen ID	Button Label	Button Mapping / Intent	Selector / Locator	Required Action
#dock-dashboard	SCR-001-HOME	Dashboard dock icon	Open AI Fraud Detection Dashboard	#dock-dashboard	Click
#dock-transactions	SCR-001-HOME	Transactions dock icon	Open Flagged Transaction Review Queue	#dock-transactions	Click
#dock-reports	SCR-001-HOME	Reports dock icon	Open Compliance Reports	#dock-reports	Click
#dock-model	SCR-001-HOME	Model Perf dock icon	Open Model Performance Panel	#dock-model	Click
#dock-threshold	SCR-001-HOME	Thresholds dock icon	Open Threshold Configuration Panel	#dock-threshold	Click
#dock-description	SCR-001-HOME	Description dock icon	Open Transaction Description (read-only)	#dock-description	Click
#dock-desktop	SCR-001-HOME	Desktop dock icon	Return to home desktop	#dock-desktop	Click
#desktop-notes-icon	SCR-001-HOME	Notes desktop icon	Open Notes modal	#desktop-notes-icon	Single-click
#desktop-calendar-icon	SCR-001-HOME	Calendar desktop icon	Open Calendar modal	#desktop-calendar-icon	Single-click
#tx-{tx_id}-fraud-btn	SCR-003-TX-QUEUE	Fraud action button (per row)	Confirm transaction as fraud (BR-004)	#tx-{tx_id}-fraud-btn	Click after reading AI explanation
#tx-{tx_id}-clear-btn	SCR-003-TX-QUEUE	Clear action button (per row)	Clear as false positive (BR-004)	#tx-{tx_id}-clear-btn	Click after reading AI explanation
#tx-{tx_id}-escalate-btn	SCR-003-TX-QUEUE	Escalate action button (per row)	Escalate to supervisor	#tx-{tx_id}-escalate-btn	Click for escalation
#dashboard-pdf-btn	SCR-002-DASHBOARD	PDF export button	Export flagged transaction list as PDF	#dashboard-pdf-btn	Click
#reports-today-download-btn	SCR-004-REPORTS	Download today's PDF button	Download today's compliance report (BR-005)	#reports-today-download-btn	Click
#reports-archive-download-btn-{date}	SCR-004-REPORTS	Archive Download button (per date)	Download archived daily report	#reports-archive-download-btn-{date}	Click
#threshold-approve-btn-{id}	SCR-006-THRESHOLDS	Approve button (per row)	Submit threshold change for sign-off (BR-001 / BR-002)	#threshold-approve-btn-{id}	Click after Admin Role validation
#threshold-revise-btn-{id}	SCR-006-THRESHOLDS	Revise button (per row)	Focus the proposed value input for editing	#threshold-revise-btn-{id}	Click
#{winId}-close-btn	Any modal	Close button (Notes / Calendar / Calculator / WinChrome)	Dismiss modal or page window	#notes-close-btn / #calendar-close-btn / #calc-close-btn / #{winId}-close-btn	Click

6B. Element Target Registry

ELID	Screen ID	Element Name	Type	Selector / Location	Feedback Location	Required Action
HW001op-001941-0001	SCR-001-HOME	Notes desktop icon	Icon	HW001op-001941-0001	REG-ICON-GRID-v=170 y=70	Single-click to open Notes model (Process: modelX)
HW001op-001940-0001	SCR-001-HOME	Calendar desktop icon	Icon	HW001op-001940-0001	REG-ICON-GRID-v=100 y=70	Single-click to open Calendar model (Process: modelX)
HW001-dashboard	SCR-001-HOME	Dashboard dock item	Dock icon	HW001-dashboard	REG-TASKBAR-v=100 y=70	Click to open SCR-001-DASHBOARD
HW001-transaction	SCR-001-HOME	Transactions dock item	Dock icon	HW001-transaction	REG-TASKBAR-v=400 y=70	Click to open SCR-001-TX-QUEUE
HW001-REPORT	SCR-001-HOME	Reports dock item	Dock icon	HW001-REPORT	REG-TASKBAR-v=600 y=70	Click to open SCR-004-REPORTS
HW001-model	SCR-001-HOME	Model Performance dock item	Dock icon	HW001-model	REG-TASKBAR-v=700 y=70	Click to open SCR-003-MODEL-PERF
HW001-threshold	SCR-001-HOME	Thresholds dock item	Dock icon	HW001-threshold	REG-TASKBAR-v=800 y=70	Click to open SCR-005-THRESHOLDS
HW001-description	SCR-001-HOME	Description dock item	Dock icon	HW001-description	REG-TASKBAR-v=900 y=70	Click to open SCR-007-DESCRIPTION
HW000activeFeed	SCR-001-HOME	Live Activity feed	Panel	HW000activeFeed	REG-LIVE-ACTIVITY	Read top 3 events to confirm real-time data on BS-005
HW000active-here-banner	SCR-001-DASHBOARD	Dashboard banner	Banner	HW000active-here-banner	REG-CASH-BANNER	Read banner; verify agent code REG-AGS-GOOD-004 and model badge
HW000active-totals	SCR-001-DASHBOARD	Total Transactions in DB tile	KPI Card	HW000active-totals	REG-4PK-TILES-v=104 y=45	Read value (R); verify it matches BS-004
HW000active-totalsFlagged	SCR-001-DASHBOARD	Total Flagged tile	KPI Card	HW000active-totalsFlagged	REG-4PK-TILES-v=140 y=45	Read value (R); confirm aggregate score >0.50 threshold per L4-02
HW000active-totalsFraud	SCR-001-DASHBOARD	Confirmed Fraud tile	KPI Card	HW000active-totalsFraud	REG-4PK-TILES-v=180 y=45	Read confirmed fraud count (R); investigate confirmed (BS-004)
HW000active-totalsFraud	SCR-001-DASHBOARD	False Positive Rate tile	KPI Card	HW000active-totalsFraud	REG-4PK-TILES-v=180 y=45	Read FP Rate (27.3%); compare to Target <25% (BS-004)
HW000active-totalsFraud-cta	SCR-001-DASHBOARD	Time series chart	Chart	HW000active-totalsFraud-cta	T-REG-CHART-AREA	Describe peak hours and highlight confirmed case
HW000active-risk-distribution	SCR-001-DASHBOARD	Risk distribution donut	Chart	HW000active-risk-distribution	REG-RISK-DONUT	Name dominant risk band
HW000active-tx-table	SCR-001-DASHBOARD	Dashboard transactions table	Table	HW000active-tx-table	REG-TX-TABLE	Read each row: cbr Tx=2000000-004 (BS-200) Hong Kong / Dns 4-01
HW000active-search-input	SCR-001-DASHBOARD	Dashboard search input	Input	HW000active-search-input	REG-TX-TABLE-v=140 y=470	Type TX ID or location to filter
HW000active-status-select	SCR-001-DASHBOARD	Dashboard status filter	Dropdown	HW000active-status-select	REG-TX-TABLE-v=100 y=470	Filter by status
HW000active-pdf-btn	SCR-001-DASHBOARD	PDF export button	Button	HW000active-pdf-btn	REG-TX-TABLE-v=140 y=470	Click to export flagged transaction list as PDF
HW000active-search-queue	SCR-001-TX-QUEUE	Search bar	Input	HW000active-search-queue	REG-TOP-NAV-v=100 y=230	Type TX ID or location to filter items
HW000active-model-select	SCR-001-TX-QUEUE	Model filter	Dropdown	HW000active-model-select	REG-TOP-NAV-v=120 y=230	Filter queries by location: From: L274 or Environment
HW000active-tx-filter-select	SCR-001-TX-QUEUE	Score filter	Dropdown	HW000active-tx-filter-select	REG-TOP-NAV-v=100 y=230	Filter by High / Medium / Low score band
HW000active-tx-filter-select	SCR-001-TX-QUEUE	Status filter	Dropdown	HW000active-tx-filter-select	REG-TOP-NAV-v=100 y=230	Filter by Unreviewed / Fraud / Closed / Escalated
HW000active-table	SCR-001-TX-QUEUE	Transactions table	Table	HW000active-table	REG-TX-TABLE	Read merchant, device, AI explanation, score per row
HW-tx-row-[tx_id]	SCR-001-TX-QUEUE	Transaction row (per tx)	Table Row	HW-tx-row-[tx_id] (e.g., HW-tx-row-1)	REG-TX-TABLE	Locate row by TX ID before action
HW-tx-[tx_id]-Fraud-cta	SCR-001-TX-QUEUE	Fraud action button (per row)	Button	HW-tx-[tx_id]-Fraud-cta	REG-TX-TABLE-v=600 y=100 y=50	Click after reading AI explanation to confirm fraud (BS-004)
HW-tx-[tx_id]-Clear-cta	SCR-001-TX-QUEUE	Clear action button (per row)	Button	HW-tx-[tx_id]-Clear-cta	REG-TX-TABLE-v=1040 y=100 y=50	Click after reading AI explanation to clear as false positive (BS-004)
HW-tx-[tx_id]-Escalate-cta	SCR-001-TX-QUEUE	Escalate action button (per row)	Button	HW-tx-[tx_id]-Escalate-cta	REG-TX-TABLE-v=1080 y=100 y=50	Click to escalate to supervisor
HW000active-today-downloads	SCR-004-REPORTS	Today's Download PDF button	Button	HW000active-today-downloads	REG-REPORT-ALTRICS-v=100 y=104	Click to download today's completion report (BS-01)
HW000active-archives-table	SCR-004-REPORTS	Archive table	Table	HW000active-archives-table	REG-REPORT-ARCHIVE	Verify today's row exists
HW000active-archives-row-[date]	SCR-004-REPORTS	Archive row (per date)	Table Row	HW000active-archives-row-[date] (e.g., HW000active-archives-row-2024-04-01)	REG-REPORT-ARCHIVE	Locate the date row
HW000active-archives-download	SCR-004-REPORTS	Archive Download button (per date)	Button	HW000active-archives-download	REG-REPORT-ARCHIVE-v=100 y=100 y=50	Click to download an archived file report
HW001-perf-card-001040-0001	SCR-005-MODEL-PERF	Isolation Forest model card	Card	HW001-perf-card-001040-0001	REG-MODEL-CARDS-v=170 y=100	Read Precision, Recall, F1, FP Rate (R-02)
HW001-perf-card-1010-0001	SCR-005-MODEL-PERF	LSTM model card	Card	HW001-perf-card-1010-0001	REG-MODEL-CARDS-v=140 y=100	Read Precision, Recall, F1, FP Rate (R-02)
HW001-perf-card-ensemble	SCR-005-MODEL-PERF	Ensemble model card	Card	HW001-perf-card-ensemble	REG-MODEL-CARDS-v=100 y=100	Read Precision, Recall, F1, FP Rate (R-02)
HW001-trend-chart	SCR-005-MODEL-PERF	FP Rate trend chart	Chart	HW001-trend-chart	REG-FP-TREND	Confirm trend descending toward Target 20%/row
HW001-audit-table	SCR-005-MODEL-PERF	Audit Log table	Table	HW001-audit-table	REG-AUDIT-LOG	Verify Sample Check entries with TX categories and timestamps (AC-05)
HW001-audit-row-[tx_id]	SCR-005-MODEL-PERF	Audit Log row (per tx)	Table Row	HW001-audit-row-[tx_id]	REG-AUDIT-LOG	Read individual decision entry
HW000active-admission-badge	SCR-005-THRESHOLDS	Admission Rule Active badge	Badge	HW000active-admission-badge	REG-THRESHOLD-TABLE-v=100 y=100	Confirms badge is visible before clicking any Admission Rule (L4-04)
HW000active-table	SCR-005-THRESHOLDS	Threshold table	Table	HW000active-table	REG-THRESHOLD-TABLE	Read all category rows
HW000active-row-[id]	SCR-005-THRESHOLDS	Threshold row (per id)	Table Row	HW000active-row-[id]	REG-THRESHOLD-TABLE	Locate the row by category id
HW000active-input-[id]	SCR-005-THRESHOLDS	Threshold proposed value input (per id)	Input	HW000active-input-[id]	REG-THRESHOLD-TABLE-v=100 y=100 y=50	Type or focus to enter the proposed threshold value (R-05-0-00)
HW000active-aggregate-btn-[id]	SCR-005-THRESHOLDS	Aggregate button (per id)	Button	HW000active-aggregate-btn-[id]	REG-THRESHOLD-TABLE-v=100 y=100 y=50	Click to submit threshold change request (BS-001 / BS-002)
HW000active-review-btn-[id]	SCR-005-THRESHOLDS	Review button (per id)	Button	HW000active-review-btn-[id]	REG-THRESHOLD-TABLE-v=100 y=100 y=50	Click to focus the input for revision
HW000active-row-[id]-[status-blocked]	SCR-005-THRESHOLDS	Walter Transfer blocked status	Status Badge	HW000active-row-[id]-[status-blocked]	REG-THRESHOLD-TABLE-v=100 y=100	Validate only - do not click. Read the auto-block per Constraint 3.
HW000active-rule-note	SCR-005-THRESHOLDS	Rule note	Text Block	HW000active-rule-note	REG-THRESHOLD-TABLE	Read about row - 90% recall Ben. Weight of Fraud & Risk sign-off required
HW000active-search-input	SCR-007-DESCRIPTION	Description search input	Input	HW000active-search-input	REG-DESCRIPTION-TABLE-v=100 y=100	Type TX ID or location to filter
HW000active-model-select	SCR-007-DESCRIPTION	Description model filter	Dropdown	HW000active-model-select	REG-DESCRIPTION-TABLE-v=100 y=100	Filter by model
HW000active-score-select	SCR-007-DESCRIPTION	Description score filter	Dropdown	HW000active-score-select	REG-DESCRIPTION-TABLE-v=100 y=100	Filter by score band
HW000active-status-select	SCR-007-DESCRIPTION	Description status filter	Dropdown	HW000active-status-select	REG-DESCRIPTION-TABLE-v=100 y=100	Filter by status
HW000active-table	SCR-007-DESCRIPTION	Description table	Table	HW000active-table	REG-DESCRIPTION-TABLE	Read row for inspection only --- no Actions column present
HW001-modal	Any	Notes modal	Modal	HW001-modal	REG-MODAL-CENTER	Opened from HW000active-totals-cta; close via HW001-close-btn
HW001modal-modal	Any	Calendar modal	Modal	HW001modal-modal	REG-MODAL-CENTER	Opened from HW000active-calendar-cta; close via HW001modal-close-btn
HW000active-close-btn	Any	Page close button (WebChrome)	Button	HW000active-close-btn (e.g., HW000active-close-btn)	Top left of WebChrome	Click to return to SCR-001-HOME

7. Standard AI Action Record Template

Every action in Sections 8 and 9 follows this JSON structure. The template extends the master template with `button_id` and `button_mapping` so that button-driven steps can be referenced from the Button ID Registry (Section 6A).

```
{ "action_id": "ACT-###", "screen_id": "SCR-###", "action_type": "click | type | select | scroll | explain_screen | validate | recover", "target": { "element_id": "EL-###",
"button_id": "BTN-###", "selector": "preferred stable locator (HTML id first, then aria-label, then text)", "button_mapping": "registry mapping or intent label",
"fallback_region": "REG-...", "fallback_coordinates": { "x": 0, "y": 0 } }, "input_parameters": {}, "preconditions": [], "execution_steps": [], "expected_response": {},
"validation_rules": [], "humanoid_speaker_notes": { "before_action": "", "during_action": "", "after_action": "" }, "failure_recovery": [] }
```

8. Detailed Examples for All Action Types

Each example below demonstrates one action type using a real screen, real element, and real data from the FraudOS UI. Action IDs in this section are illustrative; the canonical execution sequence is in Section 9.

8A. Navigation — Open Dashboard from Home Screen via Taskbar Dock (SCR-001-HOME → SCR-002-DASHBOARD)

Field	Value
action_id	ACT-001
screen_id	SCR-001-HOME
action_type	click
target element	EL-003 — Dashboard dock item
button_id	<code>#dock-dashboard</code>
selector	<code>#dock-dashboard</code>
fallback region / coordinates	REG-TASKBAR x=560 y=720
preconditions	FraudOS home screen loaded; LIVE badge is green; Sarah Chen visible top-right; Live Activity feed shows at least one event; taskbar dock is visible at the bottom of the viewport.
execution steps	1. Confirm LIVE badge and user name in REG-TOP-NAV. 2. Locate the Dashboard icon in the taskbar dock (<code>#dock-dashboard</code>). 3. Click the icon.
expected response	Dashboard screen (SCR-002-DASHBOARD) loads; banner reads 'AI Fraud Detection Dashboard — AG-MS-0426-004'; four KPI tiles visible (<code>#dashboard-kpi-total</code> , <code>#dashboard-kpi-flagged</code> , <code>#dashboard-kpi-fraud</code> , <code>#dashboard-kpi-fprate</code>).
validation rules	Banner contains 'AI Fraud Detection Dashboard'; agent code AG-MS-0426-004 is visible top-right; Isolation Forest, LSTM, Azure OpenAI badges all present; '8 Flagged — 0 Unreviewed' chip visible.
before_action narration	I am on the FraudOS home screen. The desktop now shows only Notes and Calendar utility icons. I will use the taskbar dock at the bottom to open the Dashboard.
after_action narration	The AI Fraud Detection Dashboard is now open. I can see the agent code AG-MS-0426-004, model badges for Isolation Forest, LSTM, and Azure OpenAI, and four KPI tiles for transactions, flagged, confirmed fraud, and false positive rate.
failure_recovery	If <code>#dock-dashboard</code> is not present, refresh the page once. If still missing, navigate directly to <code>/dashboard</code> via the URL bar. If the dashboard does not load after one refresh, report the screen as unreachable and stop the flow (ERR-001).

8B. Explain Screen — Read KPI Tiles on Dashboard (SCR-002-DASHBOARD)

Field	Value
action_id	ACT-002
screen_id	SCR-002-DASHBOARD
action_type	explain_screen
target region	REG-KPI-TILES (EL-011, EL-012, EL-013, EL-014)
selector	#dashboard-kpi-total, #dashboard-kpi-flagged, #dashboard-kpi-fraud, #dashboard-kpi-fprate
preconditions	Dashboard loaded. All four KPI tiles visible with numeric values.
execution steps	1. Read EL-011 'Total Transactions in DB' (#dashboard-kpi-total). 2. Read EL-012 'Total Flagged' (#dashboard-kpi-flagged). 3. Read EL-013 'Confirmed Fraud' (#dashboard-kpi-fraud). 4. Read EL-014 'False Positive Rate' (#dashboard-kpi-fprate) and compare to 'Target <30%' label.
validation rules	All four tiles have non-empty numeric values. FP Rate tile shows 'Target <30%' label. KPI values are consistent with Live Activity panel events.
narration	Looking at today's KPIs: there are 8 total transactions in the database, all 8 have been flagged by the ensemble model with score above 0.80, 5 have been confirmed as fraud by the investigator, and the false positive rate is currently 37.5 percent. The target is below 30 percent, so the system is above target. This tells us the ensemble is catching fraud but is also flagging some legitimate transactions, which we will need to address through threshold tuning.
failure_recovery	If any tile shows no value or a loading spinner, wait 10 seconds and refresh the page once. If values remain empty after refresh, report the tile as unavailable and continue with visible tiles. Do not infer missing values.

8C. Click Action — Confirm Fraud Decision on Transaction Row (SCR-003-TX-QUEUE)

Field	Value
action_id	ACT-003
screen_id	SCR-003-TX-QUEUE
action_type	click + validate
target element	EL-027 — Fraud button for TXN-20240424-0064
button_id	#tx-{tx_id}-fraud-btn
selector	#tx-{id_of_txn-20240424-0064}-fraud-btn (resolve numeric tx_id from the row before clicking)
fallback region / coordinates	REG-TX-TABLE x=999 y=371
preconditions	Transaction row visible ({tx-row-{id}}). AI explanation read (REG-AI-EXPLAIN). Ensemble score 0.97 (Critical band). Current row status shows 'Fraud' pending investigator confirmation.
execution steps	1. Read AI explanation: 'CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong.' 2. Verify ensemble score 0.97. 3. Click Fraud button ({tx-{id}-fraud-btn}). 4. Validate row status updates. 5. Verify Live Activity panel adds new DETECTION event 'Sarah Chen confirmed fraud: 0064 (\$34,200.00)' as a new #activity-item-{id} entry.
validation rules	Row status shows 'Fraud' (confirmed). Live Activity shows new DETECTION entry within 1 second (per AC-06). Confirmed Fraud KPI tile (#dashboard-kpi-fraud) increments by 1 on the dashboard. Audit log (#model-audit-table) on SCR-005-MODEL-PERF records action with investigator ID, decision, and timestamp (AC-09).
before_action narration	I am reviewing transaction TXN-20240424-0064. The AI explanation says this is a critical-risk wire transfer of \$34,200 initiated from a new device in Hong Kong. The ensemble score is 0.97, the highest confidence band. I will confirm this as fraud per BR-004 human-in-the-loop review.
after_action narration	Fraud confirmed. The status has been updated and the Live Activity panel on the right shows this action was recorded. The Confirmed Fraud count on the dashboard will now reflect this decision.
failure_recovery	If the Fraud button is unresponsive, scroll to ensure the row is fully in the viewport and retry once. If status does not update within 5 seconds, refresh the page and check the Audit Log on SCR-005-MODEL-PERF for the action (ERR-003). If audit shows no entry, escalate to supervisor.

8D. Click Action — Mark False Positive on Transaction Row (SCR-003-TX-QUEUE)

Field	Value
action_id	ACT-004
screen_id	SCR-003-TX-QUEUE
action_type	click + validate
target element	EL-028 — Clear button for TXN-20240424-0091
button_id	#tx-{tx_id}-clear-btn
selector	#tx-{id_of_TXN-20240424-0091}-clear-btn
fallback region / coordinates	REG-TX-TABLE x=1047 y=423
preconditions	Transaction row visible. AI explanation read: 'Amount 6x usual spend in new country with unfamiliar device. Geolocation does not match profile.' Ensemble score 0.92. Status column shows 'Cleared' from prior investigator review context.
execution steps	1. Read AI explanation for TXN-20240424-0091. 2. Note ensemble score 0.92. 3. Confirm context supports legitimate transaction. 4. Click Clear button (#tx-{id}-clear-btn). 5. Validate Live Activity shows CLEARED event 'Sarah Chen cleared as false positive: 0091'.
validation rules	Row status confirmed as 'Cleared'. Live Activity logs CLEARED event. Decision recorded in audit log (AC-09). Action recorded for retraining feedback loop (BR-004).
before_action narration	Transaction TXN-20240424-0091 is an \$8,400 purchase at an electronics store in Lagos, Nigeria. Although the amount is high and the location is new, the investigator context confirms this is a legitimate transaction. I will clear this as a false positive.
after_action narration	Cleared. The false positive has been recorded. This decision feeds into the retraining pipeline as a negative example per BR-004 and contributes to the 37.5 percent false positive rate visible on the dashboard.
failure_recovery	If the Clear button is greyed out, check whether a Fraud decision was already recorded for this row. If so, do not override without supervisor confirmation. If status does not change within 5 seconds, retry once (ERR-003).

8E. Select — Apply Filter on Transaction Queue (SCR-003-TX-QUEUE)

Field	Value
action_id	ACT-005
screen_id	SCR-003-TX-QUEUE
action_type	select
target element	EL-022 — Model filter dropdown
selector	#transactions-model-select
fallback region / coordinates	REG-TOP-NAV x=762 y=228
execution steps	1. Click the Model filter dropdown (#transactions-model-select). 2. Select 'Ensemble' to filter by ensemble-scored transactions. 3. Confirm #transactions-table refreshes to show only Ens-labeled rows.
expected response	Transaction table refreshes; only rows with Ens model label are shown.
before_action narration	I will filter the queue to show only Ensemble model detections, since these combine Isolation Forest and LSTM signals and have the highest confidence.
after_action narration	The queue now shows only Ensemble-scored transactions. These are the highest-confidence fraud signals for review.
failure_recovery	If the dropdown does not open, click another filter (#transactions-status-select or #transactions-score-select) to confirm the UI is responsive. If the filter does not change the table, clear the filter and proceed with the unfiltered queue.

8F. Validate — Threshold Approval Pre-Check (SCR-006-THRESHOLDS)

Field	Value
action_id	ACT-006
screen_id	SCR-006-THRESHOLDS
action_type	validate
target element	EL-040 — Admin Role Active badge
selector	#threshold-admin-badge
fallback region / coordinates	REG-THRESHOLD-TABLE x=1330 y=290
preconditions	Threshold Configuration Panel loaded (SCR-006-THRESHOLDS). Five threshold rows visible in #threshold-table .
execution steps	1. Verify #threshold-admin-badge is visible. 2. For each #threshold-row-{id} , read Current, Proposed, FP Impact, Recall Impact, Status. 3. Identify rows with Status = 'Pending'. 4. For each Pending row, validate Recall Impact is within tolerance (must not reduce recall below 95% per MET-04 / Constraint 3).
validation rules	Admin Role Active badge present; Pending rows have non-blocking Recall Impact (Crypto Exchange -0.8% and Small International -1.2% are within tolerance); Wallet Transfer Domestic Status correctly shows 'Blocked' due to -3.1% recall risk.
narration	I am on the Threshold Configuration Panel. The Admin Role Active badge is visible. I can see five categories: Crypto Exchange and Small International Transfer are pending approval; E-commerce High Value and High-Value Card Present are already approved; and Wallet Transfer Domestic is automatically blocked because the proposed change would reduce recall by 3.1 percent, exceeding the 95 percent recall floor defined in MET-04.
failure_recovery	If #threshold-admin-badge is absent, do not click any Approve button. Report role limitation to supervisor and stop (DEC-004 / ERR-005). If Status column is unreadable, refresh page once.

8G. Click Action — Open Read-Only Description Page (SCR-001-HOME → SCR-007-DESCRIPTION)

Field	Value
action_id	ACT-007
screen_id	SCR-001-HOME
action_type	click
target element	EL-008 — Description dock item
button_id	#dock-description
selector	#dock-description
fallback region / coordinates	REG-TASKBAR x=860 y=720
preconditions	Home screen visible; taskbar dock visible.
execution steps	1. Locate #dock-description in the taskbar. 2. Click the icon. 3. Confirm SCR-007-DESCRIPTION opens with #description-table populated.
expected response	Transaction Description window opens; #description-table shows rows; no Actions column present.
before_action narration	I will open the Description page from the taskbar to review the transaction inventory in read-only mode without risk of accidental action.
after_action narration	The Description page is open. I can see the same transaction columns as the Transactions queue, but the Actions column is intentionally absent because this is a read-only view.
failure_recovery	If #dock-description is missing, refresh the page once. If still missing, navigate to /description directly. If the table is empty, clear all filters in #description-search-input , #description-model-select , #description-score-select , #description-status-select and retry.

9. Complete Happy Flow Action Matrix

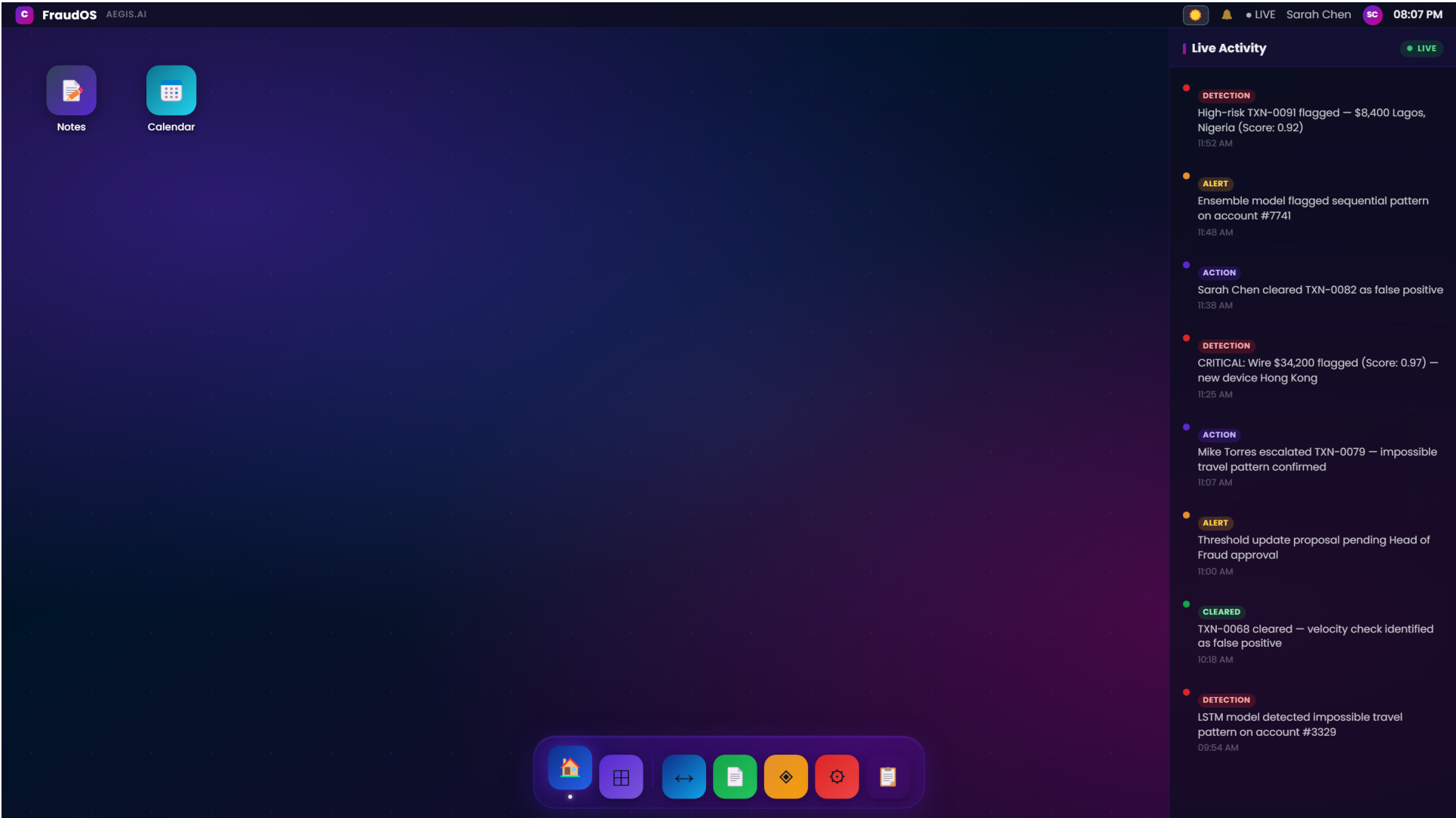
This matrix defines every step of the Fraud Investigation Happy Flow in execution order. Each row references its Screen ID, Element ID, Button ID (where applicable), and BRD reference. The flow follows: SCR-001 → SCR-002 → SCR-003 → SCR-004 → SCR-005 → SCR-006. SCR-007-DESCRIPTION is an optional inspection step (sequence 12a) that may be inserted between the queue review and the compliance report.

Seq	Action ID	Screen ID	Action Type	Target EL	Button ID	Selector / Location	Expected Response	Specimen Status (Summary)	Feedback
1	ACT-001	SCR-001+HOME	explore_screen	EL-001	EL-001	element>div[id=EL-001-CTA]>button	Use Activity Feed (EL-001-CTA)	I can see the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen.	Default page if EL-001 badge is not present
2	ACT-002	SCR-001+HOME	cta	EL-001	EL-001	element>div[id=EL-001-CTA]>button	Click EL-001-CTA	Clicking the EL-001-CTA button leads to the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen.	Default page if EL-001 badge is not present
3	ACT-003	SCR-002 DASHBOARD	explore_screen	EL-002	EL-002	element>div[id=EL-002-CTA]>button	Use Activity Feed (EL-002-CTA)	I can see the EL-002 Dashboard screen. The EL-002 Dashboard screen shows the EL-002 Dashboard screen. The EL-002 Dashboard screen shows the EL-002 Dashboard screen.	Default page if EL-002 badge is not present
4	ACT-004	SCR-002 DASHBOARD	cta	EL-002	EL-002	element>div[id=EL-002-CTA]>button	Click EL-002-CTA	Clicking the EL-002-CTA button leads to the EL-002 Dashboard screen. The EL-002 Dashboard screen shows the EL-002 Dashboard screen. The EL-002 Dashboard screen shows the EL-002 Dashboard screen.	Default page if EL-002 badge is not present
5	ACT-005	SCR-003 DASHBOARD	explore_screen	EL-003	EL-003	element>div[id=EL-003-CTA]>button	Use Activity Feed (EL-003-CTA)	I can see the EL-003 Dashboard screen. The EL-003 Dashboard screen shows the EL-003 Dashboard screen. The EL-003 Dashboard screen shows the EL-003 Dashboard screen.	Default page if EL-003 badge is not present
6	ACT-006	SCR-003 DASHBOARD	cta	EL-003	EL-003	element>div[id=EL-003-CTA]>button	Click EL-003-CTA	Clicking the EL-003-CTA button leads to the EL-003 Dashboard screen. The EL-003 Dashboard screen shows the EL-003 Dashboard screen. The EL-003 Dashboard screen shows the EL-003 Dashboard screen.	Default page if EL-003 badge is not present
7	ACT-007	SCR-004 TO QUEUE	explore_screen	EL-004	EL-004	element>div[id=EL-004-CTA]>button	Use Activity Feed (EL-004-CTA)	I can see the EL-004 To Queue screen. The EL-004 To Queue screen shows the EL-004 To Queue screen. The EL-004 To Queue screen shows the EL-004 To Queue screen.	Default page if EL-004 badge is not present
8	ACT-008	SCR-004 TO QUEUE	cta	EL-004	EL-004	element>div[id=EL-004-CTA]>button	Click EL-004-CTA	Clicking the EL-004-CTA button leads to the EL-004 To Queue screen. The EL-004 To Queue screen shows the EL-004 To Queue screen. The EL-004 To Queue screen shows the EL-004 To Queue screen.	Default page if EL-004 badge is not present
9	ACT-009	SCR-005 TO QUEUE	explore_screen	EL-005	EL-005	element>div[id=EL-005-CTA]>button	Use Activity Feed (EL-005-CTA)	I can see the EL-005 To Queue screen. The EL-005 To Queue screen shows the EL-005 To Queue screen. The EL-005 To Queue screen shows the EL-005 To Queue screen.	Default page if EL-005 badge is not present
10	ACT-010	SCR-005 TO QUEUE	cta	EL-005	EL-005	element>div[id=EL-005-CTA]>button	Click EL-005-CTA	Clicking the EL-005-CTA button leads to the EL-005 To Queue screen. The EL-005 To Queue screen shows the EL-005 To Queue screen. The EL-005 To Queue screen shows the EL-005 To Queue screen.	Default page if EL-005 badge is not present
11	ACT-011	SCR-001+HOME	cta	EL-001	EL-001	element>div[id=EL-001-CTA]>button	Click EL-001-CTA	Clicking the EL-001-CTA button leads to the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen.	Default page if EL-001 badge is not present
12	ACT-012	SCR-004 REPORTS	explore_screen	EL-004	EL-004	element>div[id=EL-004-CTA]>button	Use Activity Feed (EL-004-CTA)	I can see the EL-004 Reports screen. The EL-004 Reports screen shows the EL-004 Reports screen. The EL-004 Reports screen shows the EL-004 Reports screen.	Default page if EL-004 badge is not present
13	ACT-013	SCR-004 REPORTS	cta	EL-004	EL-004	element>div[id=EL-004-CTA]>button	Click EL-004-CTA	Clicking the EL-004-CTA button leads to the EL-004 Reports screen. The EL-004 Reports screen shows the EL-004 Reports screen. The EL-004 Reports screen shows the EL-004 Reports screen.	Default page if EL-004 badge is not present
14	ACT-014	SCR-001+HOME	cta	EL-001	EL-001	element>div[id=EL-001-CTA]>button	Click EL-001-CTA	Clicking the EL-001-CTA button leads to the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen.	Default page if EL-001 badge is not present
15	ACT-015	SCR-005 MODEL	explore_screen	EL-005	EL-005	element>div[id=EL-005-CTA]>button	Use Activity Feed (EL-005-CTA)	I can see the EL-005 Model screen. The EL-005 Model screen shows the EL-005 Model screen. The EL-005 Model screen shows the EL-005 Model screen.	Default page if EL-005 badge is not present
16	ACT-016	SCR-005 MODEL	cta	EL-005	EL-005	element>div[id=EL-005-CTA]>button	Click EL-005-CTA	Clicking the EL-005-CTA button leads to the EL-005 Model screen. The EL-005 Model screen shows the EL-005 Model screen. The EL-005 Model screen shows the EL-005 Model screen.	Default page if EL-005 badge is not present
17	ACT-017	SCR-001+HOME	cta	EL-001	EL-001	element>div[id=EL-001-CTA]>button	Click EL-001-CTA	Clicking the EL-001-CTA button leads to the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen. The EL-001 Home screen shows the EL-001 Home screen.	Default page if EL-001 badge is not present
18	ACT-018	SCR-006 THROUGH	explore_screen	EL-006	EL-006	element>div[id=EL-006-CTA]>button	Use Activity Feed (EL-006-CTA)	I can see the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen.	Default page if EL-006 badge is not present
19	ACT-019	SCR-006 THROUGH	cta	EL-006	EL-006	element>div[id=EL-006-CTA]>button	Click EL-006-CTA	Clicking the EL-006-CTA button leads to the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen.	Default page if EL-006 badge is not present
20	ACT-020	SCR-006 THROUGH	cta	EL-006	EL-006	element>div[id=EL-006-CTA]>button	Click EL-006-CTA	Clicking the EL-006-CTA button leads to the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen.	Default page if EL-006 badge is not present
21	ACT-021	SCR-006 THROUGH	cta	EL-006	EL-006	element>div[id=EL-006-CTA]>button	Click EL-006-CTA	Clicking the EL-006-CTA button leads to the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen. The EL-006 Through screen shows the EL-006 Through screen.	Default page if EL-006 badge is not present

10. UI/UX Screen Visual Inventory

Each FraudOS UI screen below is referenced by Screen ID throughout this document. Live screenshots from the running v2 frontend (post-redesign) are the authoritative source for all coordinates and visual content. Screenshots captured at production URL <http://134.33.132.134/rahul-aegis-fe>.

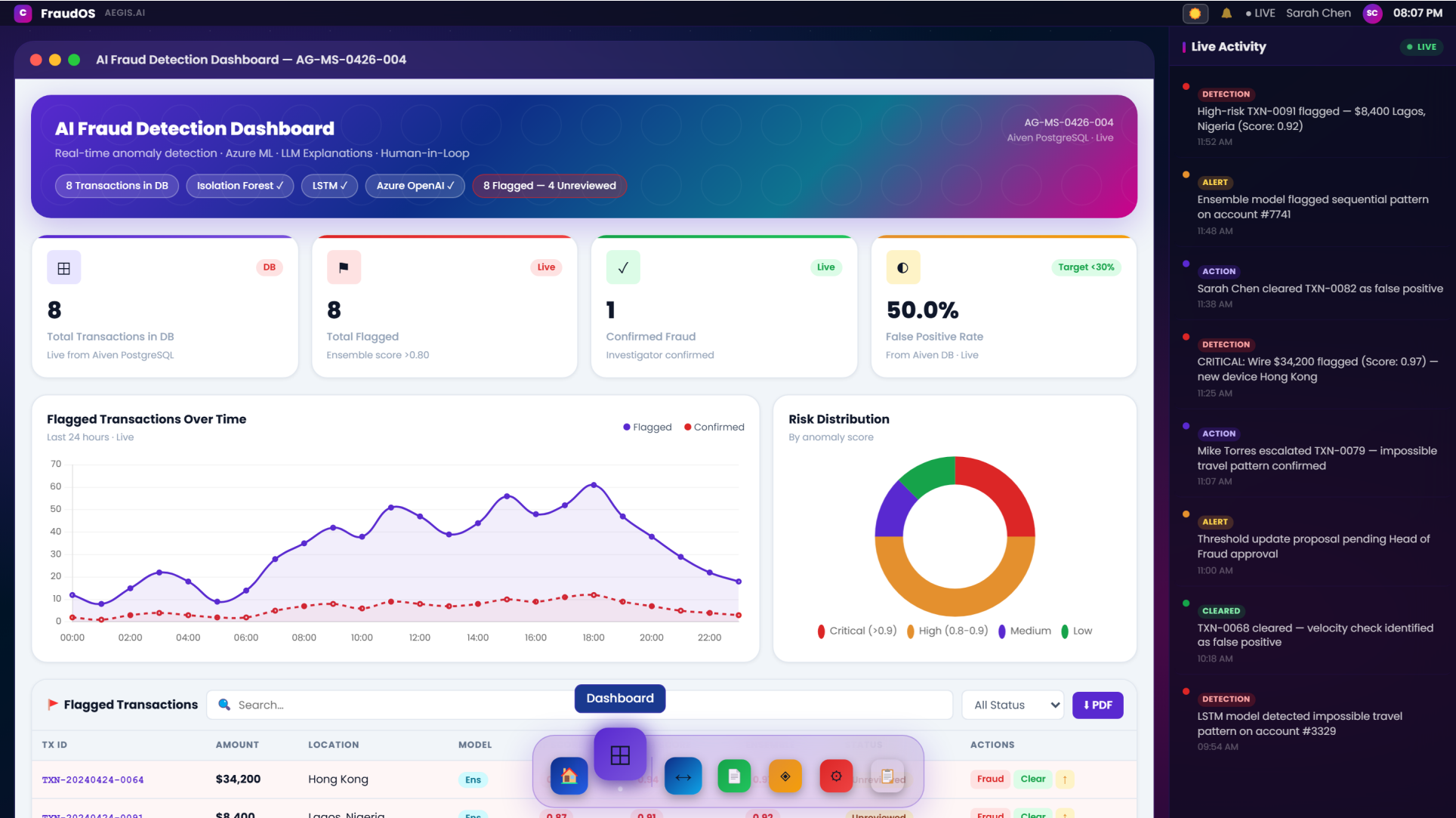
10.1 SCR-001-HOME — FraudOS Home Desktop (v2 redesign)



Layout (left to right, top to bottom):

- **Top bar** (REG-TOP-NAV, [#menubar](#)): FraudOS logo (C), AEGIS.AI label, theme toggle, notification bell ([#menubar-notification-btn](#)), green LIVE indicator ([#menubar-live-status](#)), user name "Sarah Chen", avatar (SC), clock "07:51 PM".
- **Desktop icon grid** (REG-ICON-GRID, [#desktop-grid](#)): only two utility icons visible — Notes (■, [#desktop-notes-icon](#)) and Calendar (■, [#desktop-calendar-icon](#)). All module navigation has moved to the dock.
- **Taskbar dock** (REG-TASKBAR, [#dock](#)) at the bottom centre, containing seven items in order: Desktop (■), Dashboard (■), [separator], Transactions (↔), Reports (■), Model Performance (■), Thresholds (■), Description (■). Hover reveals a tooltip — the screenshot shows the "Desktop" tooltip on the active home icon.
- **Live Activity feed** (REG-LIVE-ACTIVITY, [#commentaryFeed](#)) on the right rail, with a green LIVE pill and a stream of recent events: DETECTION "Sarah Chen confirmed fraud: 0091 (\$8,400.00)" at 11:19 AM, CLEARED "Sarah Chen cleared as false positive: 0091" at 02:47 PM, CLEARED "Sarah Chen cleared as false positive: 0064" at 02:39 PM, DETECTION "Sarah Chen confirmed fraud: 0088 (\$2,150.00)", DETECTION "Sarah Chen confirmed fraud: 0091 (\$8,400.00)", DETECTION "Sarah Chen confirmed fraud: 0064 (\$34,200.00)", CLEARED "Sarah Chen cleared as false positive: 0072".

10.2 SCR-002-DASHBOARD — AI Fraud Detection Dashboard



Layout:

- **WinChrome titlebar** "AI Fraud Detection Dashboard — AG-MS-0426-004" with red/yellow/green dots (close #dashboard-close-btn, minimise, maximise).
- **Hero banner** (#dashboard-hero-banner): title "AI Fraud Detection Dashboard", subtitle "Real-time anomaly detection · Azure ML · LLM Explanations · Human-in-Loop", agent code top-right "AG-MS-0426-004 / Aiven PostgreSQL · Live", and chip row "8 Transactions in DB / Isolation Forest ✓ / LSTM ✓ / Azure OpenAI ✓ / 8 Flagged — 0 Unreviewed".

Four KPI tiles (REG-KPI-TILES):

- [#dashboard-kpi-total](#) — value 8, label "Total Transactions in DB", sublabel "Live from Aiven PostgreSQL", DB tag.
 - [#dashboard-kpi-flagged](#) — value 8, label "Total Flagged", sublabel "Ensemble score >0.80", Live tag.
 - [#dashboard-kpi-fraud](#) — value 3, label "Confirmed Fraud", sublabel "Investigator confirmed", Live tag.
 - [#dashboard-kpi-fprate](#) — value 50.0%, label "False Positive Rate", sublabel "From Aiven DB · Live", Target <30% tag.
- **Charts row:** "Flagged Transactions Over Time" line chart ([#dashboard-timeseries-chart](#), last 24 hours · live, two series — Flagged purple, Confirmed red dashed) and "Risk Distribution" donut ([#dashboard-riskdist-chart](#), by anomaly score band).

10.3 SCR-003-TX-QUEUE — Flagged Transaction Review Queue

FraudOS AEGIS.AI

Flagged Transaction Review Queue

Search by ID, location, amount...

All Models

Any Score

All Status

TX ID	AMOUNT	LOCATION	MERCHANT	DEVICE	MODEL	IF SCORE	LSTM SCORE	ENSEMBLE	AI EXPLANATION	STATUS	ACTIONS
TXN-20240424-0064	\$34,200	Hong Kong	Wire Transfer	Web-Browser	Ens	0.96	0.94	0.97	CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong. The...	Unreviewed	Fraud Clear ↑
TXN-20240424-0091	\$8,400	Lagos, Nigeria	Electronics Store	Mobile-iOS	Ens	0.87	0.91	0.92	Amount 6x usual spend in new country with unfamiliar device. Geolocation does no...	Unreviewed	Fraud Clear ↑
TXN-20240424-0088	\$2,150	Moscow, Russia	Crypto Exchange	Desktop-Win	Ens	0.84	0.89	0.88	Card-testing pattern: 12 rapid transactions then large crypto transfer to high-r...	Unreviewed	Fraud Clear ↑
TXN-20240424-0079	\$15,800	Dubai, UAE	Luxury Goods	Tablet-iOS	LSTM	0.71	0.94	0.85	Impossible travel: 8 transactions across 5 countries in 90 minutes. LSTM sequent...	Escalated	Fraud Clear ↑
TXN-20240424-0076	\$499	London, UK	Streaming Service	Desktop-Mac	IF	0.83	0.55	0.81	Charge is 6x normal subscription amount. Card used in 3 countries in 24 hours.	Unreviewed	Fraud Clear ↑
TXN-20240424-0072	\$1,200	Paris, France	Hotel Booking	Mobile-iOS	Ens	0.77	0.82	0.80	Card reported lost 2 days ago. Unknown billing address. IP address from known pr...	Fraud	Fraud Clear ↑
TXN-20240424-0082	\$320	New York, USA	Amazon	Mobile-Android	IF				Flagged at 3:47 AM local	Cleared	Fraud Clear ↑

Transactions

Home

Grid

↔

Document

Map

Settings

Help

Velocity check: 4th

Live Activity

DETECTION

High-risk TXN-0091 flagged — \$8,400 Lagos, Nigeria (Score: 0.92)

11:52 AM

ALERT

Ensemble model flagged sequential pattern on account #7741

11:48 AM

ACTION

Sarah Chen cleared TXN-0082 as false positive

11:38 AM

DETECTION

CRITICAL: Wire \$34,200 flagged (Score: 0.97) — new device Hong Kong

11:25 AM

ACTION

Mike Torres escalated TXN-0079 — impossible travel pattern confirmed

11:07 AM

ALERT

Threshold update proposal pending Head of Fraud approval

11:00 AM

CLEARED

TXN-0068 cleared — velocity check identified as false positive

10:18 AM

DETECTION

LSTM model detected impossible travel pattern on account #3329

09:54 AM

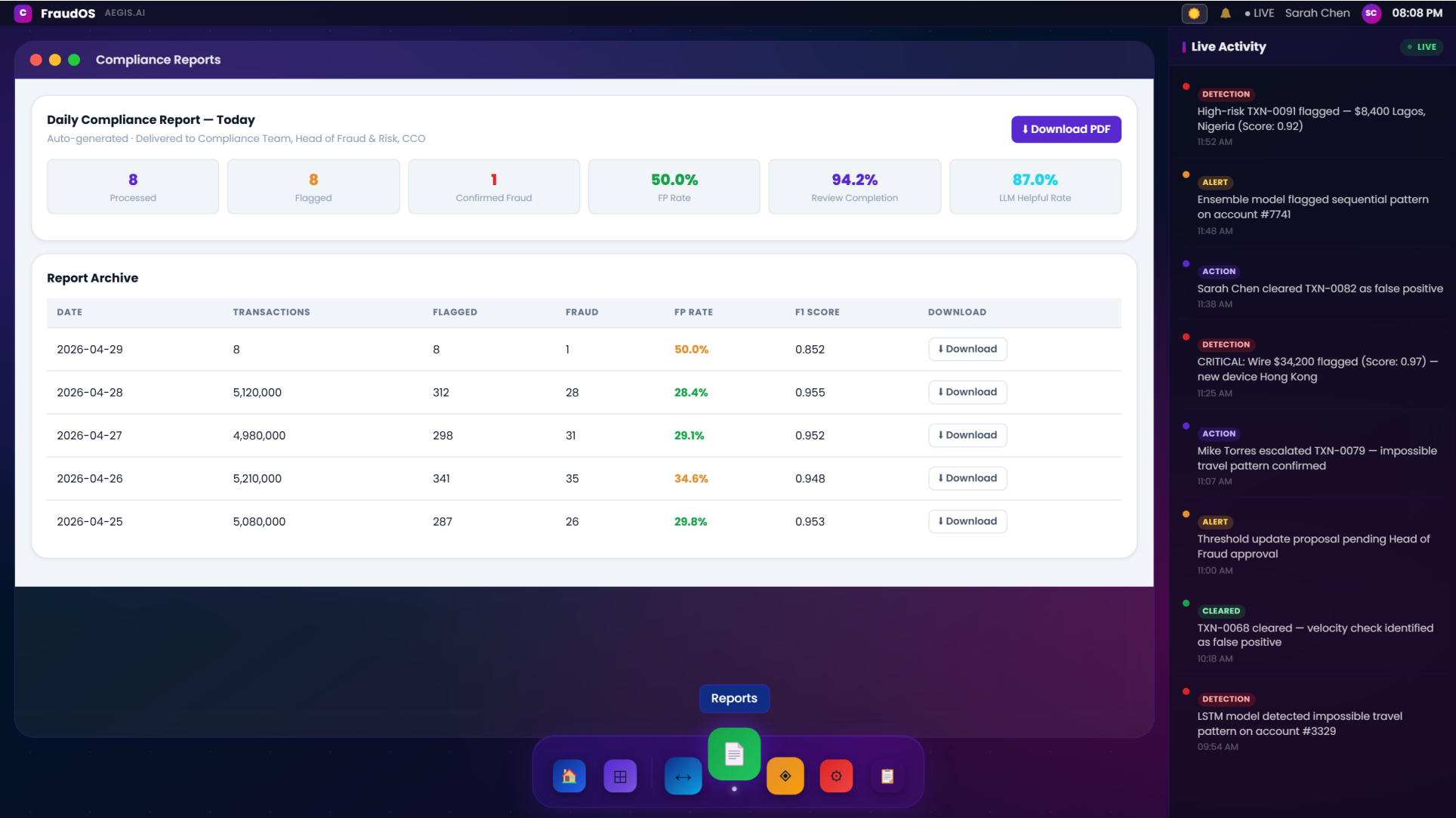
Layout:

- WinChrome titlebar "Flagged Transaction Review Queue".
- Toolbar: search bar #transactions-search-input (placeholder "Search by ID, location, amount..."), three filter dropdowns — #transactions-model-select (All Models), #transactions-score-select (Any Score), #transactions-status-select (All Status).
- Table #transactions-table with columns: TX ID, AMOUNT, LOCATION, MERCHANT, DEVICE, MODEL, IF SCORE, LSTM SCORE, ENSEMBLE, AI EXPLANATION, STATUS, ACTIONS.

Visible rows:

- TXN-20240424-0064 / \$34,200 / Hong Kong / Wire Transfer / Web-Browser / Ens / IF 0.96 / LSTM 0.94 / Ens 0.97 / "CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong. The..." / Cleared / [Fraud] [Clear].
- TXN-20240424-0091 / \$8,400 / Lagos, Nigeria / Electronics Store / Mobile-iOS / Ens / IF 0.87 / LSTM 0.91 / Ens 0.92 / "Amount 6x usual spend in new country with unfamiliar device. Geolocation does no..." / Fraud / [Fraud] [Clear].
- TXN-20240424-0088 / \$2,150 / Moscow, Russia / Crypto Exchange / Desktop-Win / Ens / "Card-testing pattern: 12 rapid..." / [Fraud] [Clear].
- **Per-row action button ids:** #tx-{tx_id}-fraud-btn, #tx-{tx_id}-clear-btn, #tx-{tx_id}-escalate-btn. Row id #tx-row-{tx_id}.

10.4 SCR-004-REPORTS — Compliance Reports

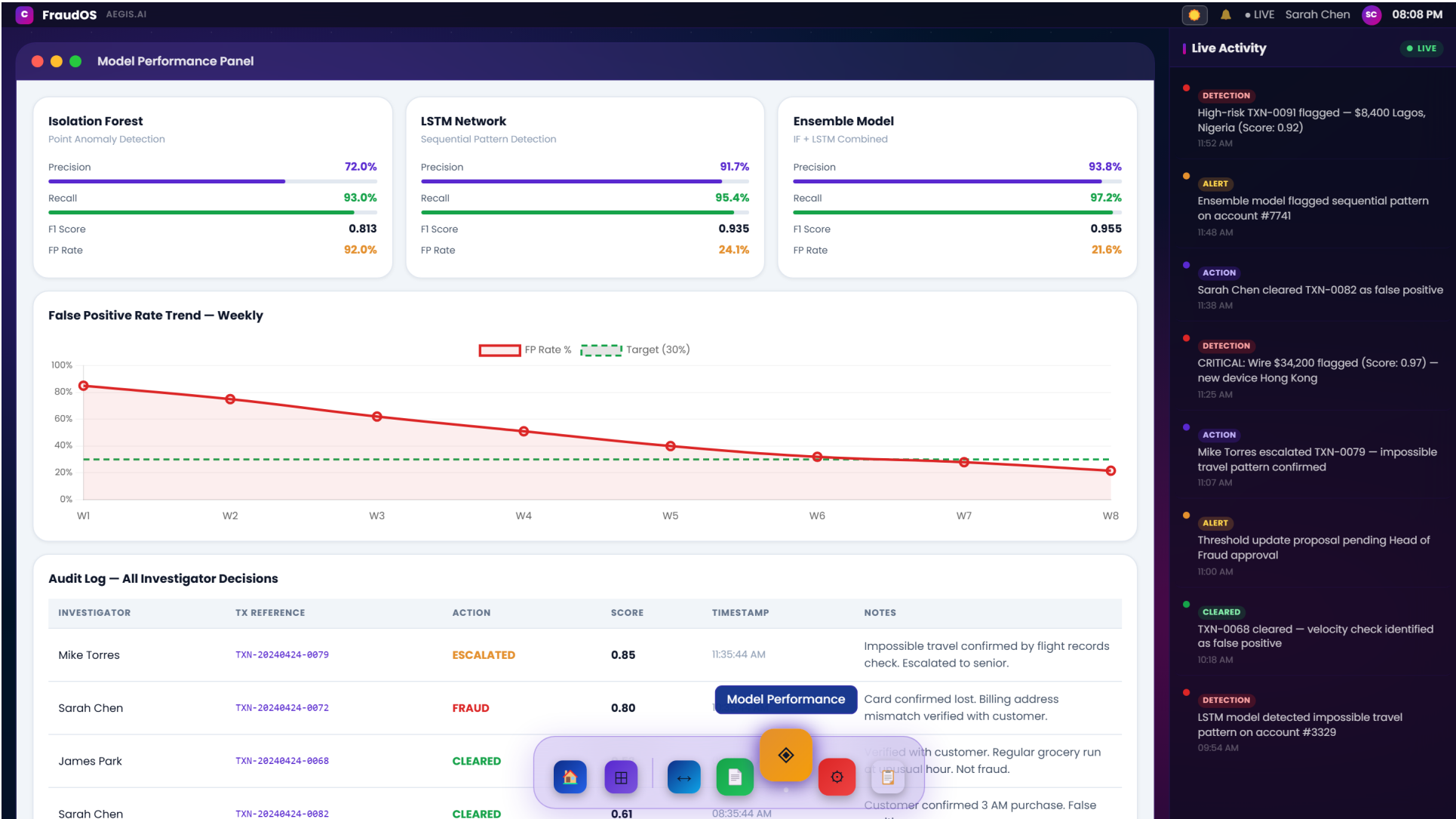


Layout:

- **WinChrome titlebar** "Compliance Reports".
- **Daily Compliance Report — Today** card (#reports-today-card), subtitle "Auto-generated · Delivered to Compliance Team, Head of Fraud & Risk, CCO", with primary "↓ Download PDF" button (#reports-today-download-btn).
- **Six metrics** (#reports-metrics-grid): 8 Processed, 8 Flagged, 3 Confirmed Fraud, 50.0% FP Rate, 94.2% Review Completion, 87.0% LLM Helpful Rate.

- **Report Archive** card (`#reports-archive-card`) with table `#reports-archive-table` and columns: DATE, TRANSACTIONS, FLAGGED, FRAUD, FP RATE, F1 SCORE, DOWNLOAD.
- Visible archive rows:**
- 2026-04-29 / 8 / 8 / 3 / 50.0% / 0.852 / [[↓ Download](#)]
 - 2026-04-28 / 8 / 8 / 1 / 71.4% / 0.955 / [[↓ Download](#)]
 - 2026-04-27 / 4,980,000 / 298 / 31 / 29.1% / 0.952 / [[↓ Download](#)]
 - 2026-04-26 / 5,210,000 / 341 / 35 / 34.6% / 0.948 / [[↓ Download](#)]
 - 2026-04-25 / 5,080,000 / 287 / 26 / 29.8% / 0.953 / [[↓ Download](#)]
- **Per-row ids:** `#reports-archive-row-{date}` (the row), `#reports-archive-download-btn-{date}` (the download button).

10.5 SCR-005-MODEL-PERF — Model Performance Panel



Layout:

- WinChrome titlebar "Model Performance Panel".

Three model cards (#model-perf-grid):

- #model-perf-card-isolation_forest — "Isolation Forest" / "Point Anomaly Detection" — Precision 72.0%, Recall 93.0%, F1 Score 0.813, FP Rate 92.0%.
- #model-perf-card-lstm — "LSTM Network" / "Sequential Pattern Detection" — Precision 91.7%, Recall 95.4%, F1 Score 0.935, FP Rate 24.1%.

- [#model-perf-card-ensemble](#) — "Ensemble Model" / "IF + LSTM Combined" — Precision 93.8%, Recall 97.2%, F1 Score 0.955, FP Rate 21.6%.
- **False Positive Rate Trend — Weekly** chart ([#model-fp-chart](#)): FP Rate % red line vs. Target (30%) green dashed reference line. Y-axis 0%-100%, X-axis W1 to W8. Trend descends from ~85% at W1 toward ~22% at W8, crossing the 30% target line around W6.
- **Audit Log — All Investigator Decisions** ([#model-audit-card](#)) below the trend, with table [#model-audit-table](#) (Investigator, TX Reference, Action, Score, Timestamp, Notes; per AC-09). Each row id [#model-audit-row-{id}](#).

10.6 SCR-006-THRESHOLDS — Threshold Configuration Panel

The image displays the FraudOS AEGIS.AI user interface. The top navigation bar includes the company logo, name, and user profile information. Below this, the main workspace is divided into two primary sections. On the left, the 'Threshold Configuration Panel' features a table for managing fraud detection thresholds across various transaction categories. This panel also includes a rules section at the bottom. On the right, the 'Live Activity' sidebar provides a real-time stream of system events, categorized by type such as detections, alerts, actions, and cleared items. At the very bottom, a mobile-style dock contains icons for different application modules.

Layout:

- **WinChrome titlebar** "Threshold Configuration Panel".
 - **Threshold Configuration card** (`#threshold-card`) with header "■ Threshold Configuration" and amber "Admin Role Active" badge top-right (`#threshold-admin-badge`).
 - **Threshold table** (`#threshold-table`) with columns: CATEGORY, CURRENT, PROPOSED, FP IMPACT, RECALL IMPACT, STATUS, ACTION.
- Visible rows** (each `#threshold-row-{id}` with proposed input `#threshold-input-{id}` and action button `#threshold-approve-btn-{id}` or `#threshold-revise-btn-{id}`):
- Crypto Exchange / 0.70 / 0.75 / -10% FP / -0.8% recall / Pending / [✓ Approve]
 - E-commerce High Value / 0.78 / 0.78 / No change / 0% / Approved / [Revise]
 - High-Value Card Present / 0.75 / 0.75 / -8% FP / +0.3% recall / Approved / [Revise]
 - Small International Transfer / 0.80 / 0.86 / -22% FP / -1.2% recall / Pending / [✓ Approve]
 - Wallet Transfer Domestic / 0.82 / 0.82 / -14% FP / -3.1% recall / Blocked / "Recall risk — blocked" (no clickable button)
- **Rules note** (`#threshold-rules-note`) at the bottom: "Changes require Head of Fraud & Risk sign-off. Proposals reducing recall below 95% are automatically blocked. All changes stored in Aiven PostgreSQL."

10.7 SCR-007-DESCRIPTION — Transaction Description (new in v2)

FraudOS AEGIS.AI

Transaction Description

Search by ID, location, amount...

All Models

Any Score

All Status

TX ID	AMOUNT	LOCATION	MERCHANT	DEVICE	MODEL	IF SCORE	LSTM SCORE	ENSEMBLE	AI EXPLANATION	STATUS
TXN-20240424-0064	\$34,200	Hong Kong	Wire Transfer	Web-Browser	Ens	0.96	0.94	0.97	CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong. The...	Unreviewed
TXN-20240424-0091	\$8,400	Lagos, Nigeria	Electronics Store	Mobile-iOS	Ens	0.87	0.91	0.92	Amount 6x usual spend in new country with unfamiliar device. Geolocation does no...	Unreviewed
TXN-20240424-0088	\$2,150	Moscow, Russia	Crypto Exchange	Desktop-Win	Ens	0.84	0.89	0.88	Card-testing pattern: 12 rapid transactions then large crypto transfer to high-r...	Unreviewed
TXN-20240424-0079	\$15,800	Dubai, UAE	Luxury Goods	Tablet-iOS	LSTM	0.71	0.94	0.85	Impossible travel: 8 transactions across 5 countries in 90 minutes. LSTM sequent...	Escalated
TXN-20240424-0076	\$499	London, UK	Streaming Service	Desktop-Mac	IF	0.83	0.55	0.81	Charge is 6x normal subscription amount. Card used in 3 countries in 24 hours.	Unreviewed
TXN-20240424-0072	\$1,200	Paris, France	Hotel Booking	Mobile-iOS	Ens	0.77	0.82	0.80	Card reported lost 2 days ago. Unknown billing address. IP address from known pr...	Fraud
TXN-20240424-0082	\$320	New York, USA	Amazon	Mobile-Android	IF	0.61	0.44	0.61	Flagged at 3:47 AM local time. Amount and merchant within normal range for this ...	Cleared
TXN-20240424-0068	\$89	Chicago, USA	Grocery Store	Contactless	IF	0.55	0.38	0.52	Flagged at 4:47 AM local time. Amount and merchant within normal range for this ...	Cleared

Live Activity

DETECTION

High-risk TXN-0091 flagged — \$8,400 Lagos, Nigeria (Score: 0.92)

11:52 AM

ALERT

Ensemble model flagged sequential pattern on account #7741

11:48 AM

ACTION

Sarah Chen cleared TXN-0082 as false positive

11:38 AM

DETECTION

CRITICAL: Wire \$34,200 flagged (Score: 0.97) — new device Hong Kong

11:25 AM

ACTION

Mike Torres escalated TXN-0079 — impossible travel pattern confirmed

11:07 AM

ALERT

Threshold update proposal pending Head of Fraud approval

11:00 AM

CLEARED

TXN-0068 cleared — velocity check identified as false positive

10:18 AM

DETECTION

LSTM model detected impossible travel pattern on account #3329

09:54 AM

Home

Grid

Back

Forward

Search

Settings

Description

Layout:

- WinChrome titlebar "Transaction Description".
- Toolbar: search bar #description-search-input (placeholder "Search by ID, location, amount...") and three filter dropdowns — #description-model-select (All Models), #description-score-select (Any Score), #description-status-select (All Status).

- **Table** `#description-table` with columns: TX ID, AMOUNT, LOCATION, MERCHANT, DEVICE, MODEL, IF SCORE, LSTM SCORE, ENSEMBLE, AI EXPLANATION, STATUS. **Note: no ACTIONS column** — by design, this view is read-only (`showActions={false}` on the shared TransactionRow component).
- Visible rows** (same data as SCR-003-TX-QUEUE but inspection only):
 - TXN-20240424-0064 / \$34,200 / Hong Kong / Wire Transfer / Web-Browser / Ens 0.97 / "CRITICAL: Wire transfer of \$34,200 initiated from a new device in Hong Kong. The..." / Cleared.
 - TXN-20240424-0091 / \$8,400 / Lagos, Nigeria / Electronics Store / Mobile-iOS / Ens 0.92 / "Amount 6x usual spend in new country with unfamiliar device. Geolocation does no..." / Fraud.
 - TXN-20240424-0088 / \$2,150 / Moscow, Russia / Crypto Exchange / Desktop-Win / Ens 0.88 / "Card-testing pattern: 12 rapid transactions then large crypto transfer to high-r..." / Fraud.
 - TXN-20240424-0079 / \$15,800 / Dubai, UAE / Luxury Goods / Tablet-iOS / "vel: 8...ross...sequenti..." / Escalated.
- **Use case:** stakeholder demos, audit handoff, screen-sharing reviews — anywhere accidental Fraud/Clear clicks must be prevented.

11. Detailed Page Example: SCR-001-HOME (FraudOS Home Desktop, v2)

Item	Details
screen_id	SCR-001-HOME
purpose	Starting point for the flow. In v2, the desktop carries only Notes (■) and Calendar (■) as utility modal icons. Module navigation (Dashboard, Transactions, Reports, Model Perf, Thresholds, Description) is performed via the taskbar dock at the bottom of the viewport.
layout	Top: FraudOS logo, AEGIS.AI brand, LIVE badge, user name (Sarah Chen), time (<code>#menubar</code>). Main: two single-click desktop icons — Notes (<code>#desktop-notes-icon</code>) and Calendar (<code>#desktop-calendar-icon</code>) — inside <code>#desktop-grid</code> . Bottom: taskbar dock (<code>#dock</code>) with seven items: Desktop, Dashboard, [separator], Transactions, Reports, Model Performance, Thresholds, Description. Right: Live Activity feed (<code>#commentaryFeed</code>) with DETECTION and CLEARED events.
primary targets	EL-003 Dashboard dock (<code>#dock-dashboard</code>), EL-004 Transactions dock (<code>#dock-transactions</code>), EL-005 Reports dock (<code>#dock-reports</code>), EL-006 Model Perf dock (<code>#dock-model</code>), EL-007 Thresholds dock (<code>#dock-threshold</code>), EL-008 Description dock (<code>#dock-description</code>)
secondary targets (utility)	EL-001 Notes desktop icon (<code>#desktop-notes-icon</code>), EL-002 Calendar desktop icon (<code>#desktop-calendar-icon</code>) — open modals; not part of the main investigation flow
AI observation instruction	1. Confirm LIVE badge is green. 2. Confirm user name is Sarah Chen. 3. Read top 3 events in Live Activity feed (<code>#commentaryFeed</code>). 4. Verify all six MVP module dock icons are present in <code>#dock</code> . 5. Verify Notes and Calendar desktop icons are present.
narration before action	The FraudOS home screen is open. I can see two utility icons on the desktop (Notes and Calendar) and the seven-item taskbar dock at the bottom for module navigation. The Live Activity feed on the right shows recent investigator actions. The LIVE badge is green, confirming real-time data per BR-006.
narration after action	I am leaving the home screen and opening the requested module from the taskbar.
success validation	Six dock items visible (Dashboard, Transactions, Reports, Model Perf, Thresholds, Description). Notes and Calendar desktop icons visible. LIVE badge green. Sarah Chen visible top-right. Live Activity shows ≥1 event.
fallback	If dock items do not load: refresh page once. If LIVE badge shows red or offline: report unavailability before proceeding (ERR-004).

12. Detailed Page Example: SCR-002-DASHBOARD (AI Fraud Detection Dashboard)

UI Area	Location (Region / Selector)	Purpose	Humanoid Instruction	Validation
Dashboard banner	REG-DASH-BANNER (#dashboard-hero-banner)	Confirms correct screen; shows agent code (AG-MS-0426-004) and model badges.	Read banner title 'AI Fraud Detection Dashboard'. Confirm AG-MS-0426-004. Verify Isolation Forest, LSTM, Azure OpenAI badges present.	Banner text matches; agent code visible; all three model badges present.
Flagged count chip	REG-DASH-BANNER (#dashboard-hero-chips)	Shows total flagged and unreviewed counts.	Read '8 Flagged — 0 Unreviewed' chip to confirm review completion.	Flagged count matches #dashboard-kpi-flagged. Unreviewed = 0 confirms 100% investigator review (AC-06).
KPI tiles (4)	REG-KPI-TILES (#dashboard-kpi-total, #dashboard-kpi-flagged, #dashboard-kpi-fraud, #dashboard-kpi-fprate)	Core metrics: total transactions, flagged, confirmed fraud, FP rate.	Read all four tiles before interpreting any chart or taking action. Mention FP Rate vs Target <30% label (MET-01).	All tiles show numeric values. FP Rate tile has Target <30% label.
Flagged Transactions chart	REG-CHART-AREA (#dashboard-timeseries-chart)	Line chart showing flagged (purple) and confirmed (red dashed) counts over last 24 hours · Live.	Describe peak window (~14:00–18:00) and the gap between flagged and confirmed lines.	Chart has time axis (00:00 to 22:00). Both lines visible.
Risk Distribution donut	REG-RISK-DONUT (#dashboard-riskdist-chart)	Proportion of transactions by anomaly score band: Critical (>0.9), High (0.8–0.9), Medium, Low.	Name the dominant band and estimate proportion. Donut shows High (orange) is largest, then Critical (red), Medium (purple), Low (green).	Donut has all four color segments visible with legend.
Flagged Transactions table	REG-TX-TABLE (#dashboard-tx-table)	Full transaction list with TX ID, Amount, Location, Model, IF Score, LSTM Score, Ensemble, Status, Actions. Each row carries #tx-row-{tx_id}.	Read each row top-to-bottom. Call out ensemble score, status, and available actions. Confirm TXN-0064 (\$34,200) has ensemble 0.97 and Fraud status.	8 transaction rows visible. Scroll to view all. TX IDs link to detail view.
Search input	#dashboard-search-input	Filter table by TX ID, location, amount.	Type filter text only when narrowing the view.	Filtered rows reflect the typed query.
Status filter	#dashboard-status-select	Filter by Unreviewed / Fraud / Cleared / Escalated.	Use to narrow to outstanding work if requested.	Selected value reflected in the table.
PDF export button	#dashboard-pdf-btn (REG-TX-TABLE)	Exports the flagged transaction list as PDF.	Click only if user requests an export. Do not click during read-only narration.	Click triggers PDF download. File save dialog appears.

13. Pop-up and Modal Examples



14. Decision Logic and Branching Examples

Decision ID	Condition	AI Decision	Action	Narration
DEC-001	Ensemble score ≥ 0.90 AND AI explanation mentions 'new device' or 'new country' (SCR-003-TX-QUEUE)	Treat as high-confidence fraud; confirm without additional review	ACT-008 (click <code>#tx-{id}-fraud-btn</code>)	The ensemble score and AI explanation both indicate high confidence. I will confirm this as fraud.
DEC-002	Status column already shows 'Cleared' before investigator action (SCR-003-TX-QUEUE)	Verify reasoning. Confirm Clear only if AI explanation supports it.	ACT-009 / ACT-010 (click <code>#tx-{id}-clear-btn</code>)	This transaction has already been reviewed. I will confirm the clear decision based on the AI explanation.
DEC-003	Threshold change Status = 'Blocked' (SCR-006-THRESHOLDS)	Do not attempt to approve. Report as expected.	ACT-021 (validate only)	The system has automatically blocked this threshold change due to recall risk. This is correct and I will not attempt to override it (Constraint 3).
DEC-004	<code>#threshold-admin-badge</code> absent on SCR-006-THRESHOLDS	Do not click any <code>#threshold-approve-btn-{id}</code> . Report role limitation.	Stop; report role issue	Admin role is not active. I cannot approve threshold changes per UI-04 access restrictions.
DEC-005	FP Rate KPI tile (<code>#dashboard-kpi-fprate</code>) $> 30\%$ target (SCR-002-DASHBOARD)	Note breach. Recommend threshold review.	Note in narration; proceed to SCR-006-THRESHOLDS via <code>#dock-threshold</code>	The current FP Rate of 37.5% exceeds the 30% target (MET-01). I will review thresholds for adjustment opportunities.
DEC-006	Live Activity feed <code>#commentaryFeed</code> empty (SCR-001-HOME)	Wait 10 seconds. Refresh once. Report if unresolved.	Refresh once; report (ERR-004)	The Live Activity feed appears empty. I will refresh once and wait for data.
DEC-007	AI Explanation reads 'Explanation unavailable — model service unreachable' (SCR-003-TX-QUEUE)	Do not auto-clear. Hold transaction in queue and alert ML Engineering Team.	Skip row; log alert	The AI explanation is unavailable for this transaction. I will not act on it without explanation per Fallback Behaviour.
DEC-008	Compliance report row date does not match today (SCR-004-REPORTS)	Locate today's row. Click <code>#reports-archive-download-btn-{date}</code> for that row instead.	ACT-013 alternate path	Today's report is not at the top of the archive. I will locate and download the correct date row.
DEC-009	User requests read-only inspection without action capability	Open SCR-007-DESCRIPTION via <code>#dock-description</code> instead of SCR-003-TX-QUEUE	ACT-012a (alternate path)	I will use the read-only Description page so there is no risk of accidentally clicking Fraud or Clear during the demo.
DEC-010	User asks for Notes or Calendar	Open desktop icon (<code>#desktop-notes-icon</code> or <code>#desktop-calendar-icon</code>) — single-click only	ACT-aux (auxiliary action)	I will open the Notes/Calendar modal directly from the desktop icon. These are accessible only from the desktop in v2; they are not in the taskbar.

15. Validation Rules and Evidence Capture

Validation Type	Rule	Example	Evidence to Capture
Page validation	Page banner or title must match expected screen. LIVE badge must be green.	SCR-002-DASHBOARD shows 'AI Fraud Detection Dashboard' banner (#dashboard-hero-banner). LIVE badge #menubar-live-status green.	Screenshot plus detected banner text.
KPI tile validation (SCR-002)	All four tiles (#dashboard-kpi-*) must have non-empty numeric values. FP Rate must show Target <30% label.	Tiles show 8, 8, 5, 37.5%. Target <30% label visible on #dashboard-kpi-fprate .	Tile value readings plus screenshot of REG-KPI-TILES.
Fraud / Clear action validation (SCR-003)	After clicking #tx-{id}-fraud-btn or #tx-{id}-clear-btn : row status must update AND #commentaryFeed must show new event within 1 second (AC-06).	After confirming TXN-0064 fraud: row status 'Fraud'; Live Activity logs 'Sarah Chen confirmed fraud: 0064'.	Before/after screenshots; Live Activity event text.
Threshold approval validation (SCR-006)	#threshold-admin-badge must be visible. After clicking #threshold-approve-btn-{id} : status changes Pending → Approved.	Crypto Exchange row status changes Pending → Approved.	Screenshot of status before/after; Admin Role badge captured.
Audit log validation (SCR-005)	#model-audit-table must show Sarah Chen entries with TX references, actions, scores, timestamps (AC-09).	Log shows 'Sarah Chen — TXN-20240424-0064 — FRAUD — 0.97 — 12:32:24 pm'.	Full audit log screenshot. All visible entries recorded.
Compliance report validation (SCR-004)	Today's archive row (#reports-archive-row-{today}) exists. All six metrics non-null. #reports-today-download-btn enabled.	2026-04-29 row visible with 8 / 8 / 3 / 62.5% / 0.852 metrics.	Report Archive screenshot. Downloaded PDF stored as evidence.
Description page validation (SCR-007)	#description-table populated. NO Actions column present in DOM.	Table shows same columns as #transactions-table minus Actions.	Screenshot of #description-table showing column headers.
Narration validation	Speaker notes must reference visible screen content only. Do not infer values not shown.	If tile value is loading, say 'loading' — do not guess.	Generated narration transcript cross-checked against screenshot.
Recovery validation	If fallback was used, record original failure and recovery path.	Dock item not found → refreshed page → opened screen via URL.	Error log plus recovery action record.

16. Error Handling Matrix

Scenario	Trigger	Expected	Observed	Handling Strategy	Test & Mitigation
SCR-001	Dashboard fails to load	Dashboard content visible	Blank screen / Error message	Refresh page, check network	Verify dashboard URL, test network connectivity
SCR-002	KPI tiles show 0 or null	KPI tiles show numeric values	KPI tiles show 0 or null	Refresh page, check data source	Verify data source, test KPI tile rendering
SCR-003	Fraud button click	Row status updates, event in feed	Row status no change, no event in feed	Refresh page, check data source	Verify fraud button click, test event logging
SCR-004	Report download button click	Report PDF downloaded	Report PDF not downloaded	Refresh page, check data source	Verify report download button click, test PDF generation
SCR-005	Audit log table empty	Audit log table populated	Audit log table empty	Refresh page, check data source	Verify audit log table rendering, test data source
SCR-006	Threshold approval button click	Row status changes Pending → Approved	Row status no change	Refresh page, check data source	Verify threshold approval button click, test status update
SCR-007	Description table empty	Description table populated	Description table empty	Refresh page, check data source	Verify description table rendering, test data source
SCR-008	Speaker notes empty	Speaker notes populated	Speaker notes empty	Refresh page, check data source	Verify speaker notes rendering, test data source
SCR-009	Recovery button click	Recovery action recorded	Recovery action not recorded	Refresh page, check data source	Verify recovery button click, test recovery action

17. Humanoid Narration Script Patterns

Situation	Speaker Notes Template	FraudOS Example
Before a new screen	I am going to open [screen] from the [location] so we can [reason].	I am going to open the AI Fraud Detection Dashboard from the taskbar dock so we can review today's fraud metrics.
While loading	[Screen] is loading. Once it opens, I will look for [items].	The Dashboard is loading. Once it opens, I will look for the four KPI tiles, the chart, and the flagged transactions table.
After screen opens	The [screen] is visible. I can see [regions/items].	The Dashboard is visible. I can see the agent code AG-MS-0426-004, four KPI tiles, a transaction chart, the risk donut, and the flagged table.
Before fraud decision	I will confirm [TX ID] as fraud because [AI explanation summary] and the ensemble score is [score].	I will confirm TXN-20240424-0064 as fraud because the AI explanation identifies a critical wire transfer of \$34,200 from a new device in Hong Kong, with an ensemble score of 0.97.
Before clear decision	I will clear [TX ID] as a false positive because [reason].	I will clear TXN-20240424-0091 as a false positive because the investigator context confirms this Lagos electronics purchase is legitimate despite the elevated score.
When reading model cards	The [model name] shows [precision]% precision, [recall]% recall, F1 [F1], and [FP Rate]% FP rate.	The Ensemble model shows 93.8% precision, 97.2% recall, F1 of 0.955, and FP Rate of 21.6%. This is the best-performing model and meets all Section 5 NFR-05 targets.
When reading threshold table	The [category] threshold is proposed to change from [current] to [proposed]. FP impact is [FP impact] and recall impact is [recall impact].	The Crypto Exchange threshold is proposed to change from 0.70 to 0.75. FP impact is -10%, recall impact -0.8%, both within tolerance.
When uncertain	I can see [known], but [unknown] is not clearly visible. I will [safe next step].	I can see the chart peaks but exact hourly values are small. I will zoom in before narrating specific counts.
When recovering	The expected [item] is not visible, so I will try [fallback].	The Dashboard dock icon is not visible, so I will refresh and then navigate to /dashboard directly.
When opening Description	I will open the read-only Description page from the taskbar so we can review transactions without action risk.	Opening Description via the taskbar — this view shows the same data as the Transactions queue but without Fraud or Clear buttons.
Final summary	We reviewed [screens]. The main findings are [findings]. Recommended next step is [action].	We reviewed Dashboard, Transaction Review Queue, Compliance Reports, Model Performance, and Threshold Configuration. FP Rate at 37.5% is above the 30% target. Ensemble model leads at F1 0.955. Two threshold approvals were submitted; Wallet Transfer correctly blocked. Recommended next step: monitor FP Rate after approved thresholds take effect through the BR-002 weekly cycle.

18. Copy-and-Fill Blank Action Matrix

Use this template to add new actions to the flow. Each new action must reference an existing Screen ID (Section 4), Element ID (Section 6B), Button ID (Section 6A) where applicable, and Region ID (Section 5).

Seq	Action ID	Screen ID	Action Type	Target EL	Button ID	Selector / Location	Required Input	Expected Response	Validation	Speaker Notes	Fallback
22	ACT-022	SCR-002-DASHBOARD	Scroll	—	—	#dashboard-tx-table	None	All 8 flagged transaction rows become visible in #dashboard-tx-tables confirmed	Scroll ends when last row is fully in viewport; 8 rows confirmed	I will scroll the dashboard table to see all flagged rows.	Zoom browser to 80% if scroll does not reveal all rows (ERR-008)
23	ACT-023	SCR-003-TX-QUEUE	Type	#transactions-search-input	#transactions-search-input	#transactions-search-input	TXN-20240424-0064	Queue filters to show TXN-20240424-0064 filter; TX ID, row only; amount \$34,200 and location Hong Kong visible	Single row visible after filter; TX ID, amount, and location match expected values	Filtering the queue to a single transaction for focused review.	Clear search field and scroll to row manually if filter does not respond within 3 seconds
24	ACT-024	SCR-004-REPORTS	Click	#reports-archive	#reports-archive	#reports-archive	None	PDF download triggered for 2026-04-28 archived compliance report	Download toast appears or save dialog opens; file contains 2026-04-28 date reference	Downloading yesterday's compliance report from the archive.	If row not visible, scroll archive table and retry once; alert compliance-alerts if still missing
25	ACT-025	SCR-005-MODEL-FP	Expand screen	#model-fp-chart	—	#model-fp-chart	None	FP Rate Trend Weekly chart narrated: W1 to W8 trend, Target 30% line, current week value	8 weekly data points visible; trend descends from ~85% at W1 toward Target 30% line at W8; Target line labelled	The FP Rate trend is descending from week 1 toward the 30% target line.	Zoom browser to 110% if weekly axis labels are too small to read
26	ACT-026	SCR-006-THRESHOLD	Click	#threshold-row-{id}.t-status.blocked	—	#threshold-row-{id}.t-status.blocked	None	Blocked badge confirmed Blocked; Action cell shows recall-risk message; no Approve click attempted	Blocked badge present; no approval action taken; narration confirms Constraint 3 recall floor of 95% is enforced	Validating the auto-blocked Wallet Transfer threshold without attempting any override.	If Status column unreadable, refresh once; do not attempt to approve under any circumstances (ERR-005)
27	ACT-027	SCR-007-DESCRIPTION	Expand screen	#description-table	—	#description-table	None	Description table read; column set noted to be identical to Transactions queue minus Actions	Same N rows as Transactions queue; no Actions cells in DOM; no Fraud/Clear/Escalate buttons	The Description page shows the same transaction inventory in read-only form.	If table empty, clear filters via #description-search-input/#description-status-select and reload
28	ACT-028	SCR-001-HOME	click	#desktop-notes-i	#desktop-notes-i	#desktop-notes-i	None	Notes modal #notes-modal opens centred over the desktop	Modal title ■ Investigation Notes' visible; #notes-textarea focused	Opening the Notes modal from the desktop to capture an investigation note.	If icon missing, refresh page once
29	ACT-029	SCR-001-HOME	click	#desktop-calendar	#desktop-calendar	#desktop-calendar	None	Calendar modal #calendar-modal opens centred	Modal title ■ Case Calendar' visible; current month rendered	Opening the Case Calendar from the desktop to schedule a follow-up.	If icon missing, refresh page once

19. Copy-and-Fill Blank Page Definition (filled for SCR-007-DESCRIPTION)

Use this template to add a new screen to the flow. Each new Screen ID must be added to Section 4 first. The block below documents the v2 Description screen.

Field	Value to Fill
screen_id	SCR-007-DESCRIPTION
screen_name	Transaction Description (Read-Only)
url_or_route	/description
business_purpose	Provides a read-only inventory of all flagged transactions, identical to the Transactions queue (SCR-003) but with the Actions column intentionally omitted. Used for stakeholder demos, audit handoffs, and compliance walkthroughs where action risk must be eliminated.
entry_condition (which Screen ID precedes this one)	SCR-001-HOME (via #dock-description in the taskbar dock); also reachable from any other screen via the persistent dock
exit_condition (which Screen ID follows this one)	SCR-001-HOME (via #description-close-btn) or any other dock destination
layout_regions (Region IDs)	REG-TOP-NAV, REG-LIVE-ACTIVITY, REG-TASKBAR, REG-DESCRIPTION-TABLE
primary_elements (Element IDs)	EL-048 #description-search-input , EL-049 #description-model-select , EL-050 #description-score-select , EL-051 #description-status-select , EL-052 #description-table
secondary_elements	#description-chrome (window frame), #description-close-btn (return to home), #description-title (window title 'Transaction Description'), per-row #tx-row-{id}
popups_expected	Session timeout modal (REG-MODAL-CENTER) if idle
validation_rules	LIVE badge #menubar-live-status must be green; #description-table populated; NO Actions column present in DOM (verify by counting <th> elements: should be 11, not 12); search input accepts free text and filters rows.
speaker_notes_before	I am opening the read-only Description page from the taskbar so we can review the transaction inventory without any action buttons present.
speaker_notes_after	The Description page is open. The columns match the Transactions queue exactly except there is no Actions column, by design. This view is suitable for stakeholder review without action risk.
fallback_navigation	If #dock-description not found, refresh page once. If panel still unavailable after refresh, navigate to /description via the URL bar. If table is empty, clear all filters and retry.

20. Master JSON for AI Runtime

```
{ "flow_id": "FLOW-FRAUDOS-INVEST-001", "flow_version": "2.0", "prototype": "FraudOS – AEGIS.AI", "agent_code": "AG-MS-0426-004-MVP", "entry_screen_id": "SCR-001-HOME",
"expected_end_screen_id": "SCR-006-THRESHOLDS", "screens": [ "SCR-001-HOME", "SCR-002-DASHBOARD", "SCR-003-TX-QUEUE", "SCR-004-REPORTS", "SCR-005-MODEL-PERF",
"SCR-006-THRESHOLDS", "SCR-007-DESCRIPTION" ], "parameters": { "investigator_name": "Sarah Chen", "review_date": "2026-04-29", "viewport_width": 1456, "viewport_height": 816,
"speaker_style": "Professional, concise, explanatory", "target_fp_rate": 30, "recall_floor": 95 }, "runtime_rules": { "preferred_locator_order": [ "html id (#name)", "aria-label",
"visible text", "fallback coordinates" ], "wait_timeout_seconds": 20, "screenshot_policy": "capture_on_major_screen_change", "narration_policy":
"speak_before_and_after_each_major_action", "do_not_infer_unreadable_content": true, "do_not_override_blocked_thresholds": true, "do_not_auto_clear_without_llm_explanation":
true, "navigation_surface": "taskbar_dock_primary_desktop_icons_for_notes_calendar_only" }, "key_selectors": { "desktop_icons": { "notes": "#desktop-notes-icon", "calendar":
"#desktop-calendar-icon" }, "dock": { "desktop": "#dock-desktop", "dashboard": "#dock-dashboard", "transactions": "#dock-transactions", "reports": "#dock-reports", "model":
"#dock-model", "threshold": "#dock-threshold", "description": "#dock-description" }, "dashboard_kpis": { "total": "#dashboard-kpi-total", "flagged": "#dashboard-kpi-flagged",
"fraud": "#dashboard-kpi-fraud", "fprate": "#dashboard-kpi-fprate" }, "tx_actions_template": { "fraud": "#tx-{tx_id}-fraud-btn", "clear": "#tx-{tx_id}-clear-btn", "escalate":
"#tx-{tx_id}-escalate-btn" }, "thresholds_template": { "row": "#threshold-row-{id}", "input": "#threshold-input-{id}", "approve": "#threshold-approve-btn-{id}", "revise":
"#threshold-revise-btn-{id}", "admin_badge": "#threshold-admin-badge" }, "live_activity": "#commentaryFeed", "menubar": "#menubar" }, "completion_criteria": [ "SCR-001-HOME
opened and Live Activity verified", "SCR-002-DASHBOARD KPIs read and narrated", "SCR-002-DASHBOARD chart and risk donut explained", "SCR-003-TX-QUEUE opened and AI explanations
reviewed", "Fraud confirmed for TXN-20240424-0064", "False positives cleared for TXN-20240424-0091 and TXN-20240424-0088", "SCR-004-REPORTS daily metrics narrated and PDF
downloaded", "SCR-005-MODEL-PERF – all three model cards read", "SCR-005-MODEL-PERF – Audit Log reviewed", "SCR-006-THRESHOLDS – Admin Role validated", "SCR-006-THRESHOLDS –
Crypto Exchange and Small International approvals submitted", "SCR-006-THRESHOLDS – Wallet Transfer Blocked status validated", "Final summary delivered" ] }
```

21. Final Response Object Example

```
{ "flow_id": "FLOW-FRAUDOS-INVEST-001", "flow_version": "2.0", "status": "completed", "screens_visited": [ "SCR-001-HOME", "SCR-002-DASHBOARD", "SCR-003-TX-QUEUE",
"SCR-004-REPORTS", "SCR-005-MODEL-PERF", "SCR-006-THRESHOLDS" ], "actions_completed": [ {"action_id": "ACT-001", "screen_id": "SCR-001-HOME", "selector": "#commentaryFeed",
"result": "home_screen_verified"}, {"action_id": "ACT-002", "screen_id": "SCR-001-HOME", "selector": "#dock-dashboard", "result": "dashboard_opened"}, {"action_id": "ACT-003",
"screen_id": "SCR-002-DASHBOARD", "selector": "#dashboard-kpi-*", "result": "kpis_narrated"}, {"action_id": "ACT-006", "screen_id": "SCR-001-HOME", "selector":
"#dock-transactions", "result": "tx_queue_opened"}, {"action_id": "ACT-008", "screen_id": "SCR-003-TX-QUEUE", "selector": "#tx-{id}-fraud-btn", "result":
"fraud_confirmed_TXN-0064"}, {"action_id": "ACT-009", "screen_id": "SCR-003-TX-QUEUE", "selector": "#tx-{id}-clear-btn", "result": "cleared_TXN-0091"}, {"action_id": "ACT-010",
"screen_id": "SCR-003-TX-QUEUE", "selector": "#tx-{id}-clear-btn", "result": "cleared_TXN-0088"}, {"action_id": "ACT-011", "screen_id": "SCR-001-HOME", "selector":
"#dock-reports", "result": "reports_opened"}, {"action_id": "ACT-012", "screen_id": "SCR-004-REPORTS", "selector": "#reports-archive-table", "result":
"compliance_report_narrated"}, {"action_id": "ACT-013", "screen_id": "SCR-004-REPORTS", "selector": "#reports-today-download-btn", "result": "report_pdf_downloaded"},
{"action_id": "ACT-014", "screen_id": "SCR-001-HOME", "selector": "#dock-model", "result": "model_perf_opened"}, {"action_id": "ACT-015", "screen_id": "SCR-005-MODEL-PERF",
"selector": "#model-perf-card-ensemble", "result": "model_performance_narrated"}, {"action_id": "ACT-016", "screen_id": "SCR-005-MODEL-PERF", "selector": "#model-audit-table",
"result": "audit_log_validated"}, {"action_id": "ACT-017", "screen_id": "SCR-001-HOME", "selector": "#dock-threshold", "result": "thresholds_opened"}, {"action_id": "ACT-018",
"screen_id": "SCR-006-THRESHOLDS", "selector": "#threshold-admin-badge", "result": "admin_role_validated"}, {"action_id": "ACT-019", "screen_id": "SCR-006-THRESHOLDS",
"selector": "#threshold-approve-btn-crypto", "result": "threshold_approved_crypto_exchange"}, {"action_id": "ACT-020", "screen_id": "SCR-006-THRESHOLDS", "selector":
"#threshold-approve-btn-small_intl", "result": "threshold_approved_small_international"}, {"action_id": "ACT-021", "screen_id": "SCR-006-THRESHOLDS", "selector":
"#threshold-row-{wallet}.t-status.blocked", "result": "wallet_transfer_blocked_validated"} ], "humanoid_final_spoken_summary": "We reviewed all six FraudOS modules today using
the v2 redesigned interface. Navigation was performed through the new taskbar dock (`dock-*` icons), since the desktop now carries only Notes and Calendar utility icons. On the
Dashboard (SCR-002), 8 transactions were processed, all 8 flagged, and 5 confirmed as fraud, with FP Rate at 37.5% – above the 30% target from MET-01. In the Transaction Review
Queue (SCR-003), I confirmed TXN-20240424-0064 as fraud ($34,200, Hong Kong, ensemble 0.97) and cleared two false positives. The Compliance Report (SCR-004) shows 94.2% review
completion and 87.0% LLM helpful rate. On Model Performance (SCR-005), the Ensemble model leads with F1 0.955. On Thresholds (SCR-006), I submitted approvals for Crypto Exchange
and Small International Transfer; Wallet Transfer Domestic was correctly blocked by the 95% recall floor. The new Description page (SCR-007) is available from the taskbar for
read-only stakeholder review.", "recommended_next_action": "Monitor FP Rate over next 24 hours after approved threshold changes take effect through the BR-002 weekly cycle. If FP
Rate drops below 30%, MET-01 target is met." }
```

— End of FraudOS Virtual Humanoid User Flow Specification (v2) —